Program overview

26-Mar-2025 11:16

Year	**2024/2025**
Organization	**Applied Sciences**
Education	**Bachelor Nanobiology**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor NB 2024			
1e jaar BSc NB 2024	**1st year BSc NB 2024**		
NB1012	Biochemistry	3	
NB1016	Molecular Biology	3	
NB1022	Genetics	4	
NB1031	Introduction to studying Nanobiology	3	
NB1035	Programming Foundations	3	
NB1052	Journal Club 1	3	
NB1102	Chemistry 1	3	
NB1110	Chemistry 2	3	
NB1120	Biomolecular Programming	3	
NB1132	Biophysics	3	
NB1140	Physics 1a	4	
NB1143	Physics 1B	3	
NB1150	Lab Course Track A-1	3	
NB1151	Lab Course Track A-2	3	
NB1163	Lab Course Track B-1	3	
NB1164	Lab Course Track B-2	3	
NB1201	Analysis 1	5	
NB1206	Analysis 2	5	
NB1211	Analysis 3	3	
NB1230	Linear Algebra	3	
2e jaar BSc NB 2024	**2nd year BSc NB 2024**		
NB2011	Thermodynamics and Transport	3	
NB2022	Philosophy and Ethics	3	
NB2032	Evolutionary Developmental Biology	6	
NB2071	Physical Biology of the Cell 2	3	
NB2081	Nanotechnology	2	
NB2141	Physics 2	3	
NB2161	Bioinformatics	4,5	
NB2171	Statistics	3	
NB2181	Computational Science	3	
NB2191	Differential equations	3	
NB2214	Electronic Instrumentation	6	
NB2220	Statistical Physics	3	
NB2230	Biomolecular Structure and Function	3	
NB2240	Imaging 2	3,5	
NB2330	Imaging 1	5	
TN2545	Systems and Signals	6	
3e jaar BSc NB 2024	**3rd year BSc NB 2024**		
NB3000	Bachelor Thesis Project	20	
3e jaar BSc NB keuzevakken 2024	**3rd year BSc NB electives 2024**		
NB3014	Computational Neuroscience	2,5	
NB3015	A primer in Neuroscience	2,5	
NB3016	High Speed Scientific Computing	2,5	
NB3017	Quantum mechanics for Nanobiology 1	2,5	
NB3018	Quantum mechanics for Nanobiology 2	2,5	
NB3020	Genomics Technology in Breast Cancer Research	2,5	
NB3021	Optics and its application in Nanobiology	2,5	
NB3022	Epigenetics	2,5	
NB3023	Human Complex Genetics	2,5	
NB3024	Advanced Math Topics	2,5	
NB3026	Biomolecular Ultrasound	2,5	
NB3030	Nanobiology Independent Research	0	
TBM301A	Writing a Bachelors Thesis in English	2,5	

Year	**2024/2025**
Organization	**Applied Sciences**
Education	**Bachelor Nanobiology**

Bachelor NB 2024	
Director of Education	Dr. J. Pothof
Program Coordinator	J.C. Colgrove
Program Coordinator	D. Mevius
Contact for students	Info-BSc-NB <Info-BSc-NB@tudelft.nl>
Program Title	BSc Nanobiology
In association with the University of	Erasmus Medical Center in Rotterdam
EC Program	The programme has a study load of 180 credits. This includes the first academic year, with a study load of 60 credits, which is concluded with a binding recommendation on the continuation of studies. The second and third academic years have a combined study load of 120 credits. This phase includes a minor with a study load of 30 credits. The programme consists of Year 1: 60 EC, courses in math, physics, biology, chemistry, programming, integrated courses, practical courses and academic skills Year 2: 60 EC, courses in math, physics, biology, integrated courses, practicals and academic skills Year 3: 30 EC minor, 20 EC bachelor end project, and 10 EC of nanobiology electives. As a component of the programme, the minor includes the following variants a. Thematic minor, as approved by the university, b. Self-composed minor, as approved by the Board of Examiners. The Bachelors degree programme is concluded with a bachelor end project. This project demonstrates that the student possesses and is able to apply the knowledge, insight and skills acquired in the degree programme.
Program Goals	The programme is intended to educate students to earn a Bachelor of Science degree in Nanobiology, providing them with such a level of knowledge, insight and skills in the area of Nanobiology, that graduates can fulfil positions on the labour market at the Bachelors level. Graduates must meet the specific final attainment levels for each degree programme. The Bachelor of Science graduate of the programme Nanobiology has the aim to integrate the fields of mathematics, physics and biology. The student: 1. will have relevant, current and fundamental knowledge of mathematics, physics and biology, as well as the methods and techniques of scientific research. 2. will be able to identify related concepts in biology and physics and to apply knowledge from one field of science to another. 3. will be able to use the acquired knowledge to follow current scientific research in the fields of biology and biophysics intensively, in addition to understanding and interpreting this literature. 4. will have demonstrable experimental research skills in the fields of molecular biology and biophysics. 5. will have the required communication skills. 6. will have an understanding of the ethical issues surrounding scientific research. 7. will be aware of the need for lifelong learning and of the utility of creativity to the achievement of scientific progress.
Gives access to	Students who have successfully completed their Bachelor's degree programme and have therefore been granted the title Bachelor of Science in Nanobiology (joint degree) are admissible to the following Master of Science degree programmes: Nanobiology joint programme with TU Delft and Erasmus University Rotterdam, and to Infection and Immunity; Molecular Medicine; and Neuroscience at Erasmus University Rotterdam Access to other Master of Science degree programmes is subject to the conditions stipulated by the university that provides that degree programme. Graduates of the Bachelor programme enter many different masters programmes.
Expected prior Knowledge	Access requirement(s), applicants must meet one of these criteria: A VWO (6 years of pre-university education) certificate which includes Mathematics B, Biology, Physics and Chemistry as VWO examination subjects. A propedeutic certificate of a university degree programme (WO) A degree certificate from a University of Applied Science (HBO): Applicants with a VWO certificate: see the VWO subject cluster requirements. Applicants with a HAVO (5 year of pre university education) or MBO (intermediate vocational education) certificate must have filled all gaps in knowledge compared to the VWO level in the following subjects: Mathematics B, Biology, Physics and Chemistry. On individual basis, proof of English proficiency can be required. Other qualifications recognised by the university's Executive Board which may or may not have been obtained in the Netherlands and are seen as being at least equivalent to a VWO examination certificate holding the subjects Mathematics, Biology, Physics and Chemistry on a proper level. On the basis of a colloquium doctum (i.e. a special entrance examination). In order to sit a colloquium doctum candidates must be at least 21 years of age. All other qualifications should have a level of education equivalent to the VWO level in the subjects Mathematics B, Biology, Physics and Chemistry.

Year	**2024/2025**
Organization	**Applied Sciences**
Education	**Bachelor Nanobiology**

1e jaar BSc NB 2024	
Responsible Program Employee	J.C. Colgrove
Director of Education	Dr. J. Pothof
Program Coordinator	J.C. Colgrove
Program Coordinator	D. Mevius

NB1012	Biochemistry	3
Responsible Instructor	Dr. B.C. Tilly	
Responsible Instructor	Dr. G.E. Bokinsky	
Instructor	Dr. B.C. Tilly	
Instructor	Dr. J.A. van der Knaap	
Instructor	Dr. G.E. Bokinsky	
Contact Hours / Week x/x/x/x	0/0/0/8/0/0/0/0	
Education Period	2 2B	
Start Education	2B	
Exam Period	2 3	
Course Language	English	
Course Contents	Biochemistry represents the interface between chemistry and biology and studies life at a molecular level. Our knowledge about the molecular composition of cells and the complex network of chemical reactions that take place within them is expanding rapidly. Nowadays, biochemistry plays an important role in many major developments in healthcare, (bio-)technology and agriculture. In this course you will learn the biochemical principles that underlie cellular physiology. In addition, you will study the molecular composition of cells and the major metabolic pathways involved, with special emphasis will on the structure and function of enzymes and bioenergetics. The knowledge obtained during this course will be applied in other biologically orientated courses, including Lab course, Biophysics, Physical Biology of the Cell and Evolutionary & Developmental Biology. For successful completion of this course, you are expected to be familiar with the concepts and skills of the courses Analysis I and Chemistry I & II.	
Study Goals	At the completion of this course you will be able to: 1. describe and recognize the major molecular constituents of the cell, their function and how they are synthesized from precursors. 2. describe the modular organization of proteins, how they fold into 3D structures and can form (multifunctional) complexes. 3. predict how activators and inhibitors affect the rate of the reaction using the principles of enzymatic catalysis. 4. Calculate the rate of enzymatic reactions using Michaelis-Menten theory. 5. Predict the effects of specific mutations on enzyme catalysis. 6. know the major metabolic pathways in bacteria and cells and explain how these pathways are regulated to maintain homeostasis. 7. describe photosynthesis in bacteria and plants and how energy is stored in reduction equivalents and ATP. 8. describe how mitochondria convert the energy stored in reduction equivalents into ATP and how the ATD/ADP ratio regulates cellular metabolism.	
Education Method	This course contains lectures, tutorials and practicals. During the practicals the properties and kinetics of enzymes will be studied using computer simulations and a laboratory experiment (Practical Alkaline phosphatase, week 3). Attendance of the practical Alkaline phosphatase is mandatory.	
Literature and Study Materials	Molecular Biology of the Cell, Seventh edition, Alberts et al., W.W. Norton & Company, 2022 Achieve Macmillan learning (subscription required, provided by the program for Nanobiology students). Additional learning materials (textbook extracts) will be posted to Brightspace.	
Assessment	The final grade is based on a written exam and points earned for sufficiently completed home assignments (up to 0.5 points, maximum grade is 10). The bonus points will be added to your grade only if you pass the written exam.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Remarks	Ben Tilly can be reached at b.tilly@erasmusmc.nl	
Studyload/Week	The course (5 weeks) consists of lectures, online and in class tutorials (32h), practical work (4h), home assignments (8h) and 40h of study at home and exam preparation. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Rotterdam	

NB1016	Molecular Biology	3
Responsible Instructor	N. Taneja	
Responsible Instructor	A. Ray Chaudhuri	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	NB1016 provides basic knowledge that is required in a number of subsequent related courses, including Physical Biology of the Cell, Biophysics, and Evolutionary & Developmental Biology.	
Course Contents	Molecular Biology focuses on 1) storage and transfer of information in biological systems, from DNA to mRNA to protein, and 2) the various mechanisms by which the integrity of DNA is maintained. Throughout the course there will be specific focus on structural requirements, as well spatial and temporal aspects of the specific pathways.	
Study Goals	1. Describe the structure of DNA 2. Describe the anatomy of a DNA replication fork and function of replication proteins 3. Explain how accuracy in DNA replication is achieved 4. Explain the end replication problem 5. Describe how transcription is initiated using general transcription factors 6. Described how different classes of transcription factors interact with DNA 7. Explain how transcription can be regulated using gene regulatory proteins 8. Explain how proteins are synthesized through translation of mRNA 9. Describe how proteins are degraded by the proteosome 10. Describe at the basic level DNA repair pathways and the lesions they act on 11. Understand the link between DNA recombination and replication 12. Describe the different phases of the cell cycle 13. Understand how the transitions between the different cell cycle phases occur 14. Explain how cell cycle checkpoints are established	
Education Method	- Lectures - Individual/group assignments	
Assessment	1 exam counts for 100%. Due to yet unforeseen developments due to COVID-19 the way the course is assessed might change.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Remarks	Wim Vermuelen can be reached at w.vermeulen@erasmusmc.nl Arnab Ray Chaudhuri can be reached at a.raychaudhuri@erasmusmc.nl Nitika Taneja can be reached at n.taneja@erasmusmc.nl	
Studyload/Week	7h/week * 8 = 56 h, including lectures, problem sessions and self study 28 h exam preparation Total of 84 hours Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Rotterdam	

NB1022	Genetics	4
Responsible Instructor	Dr. W.J. Lans	
Responsible Instructor	Dr.ir. J.A.F. Marteijn	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course the student will learn the principles of genetics, ranging from classical Mendelian genetics to the most recent molecular genetics techniques. Both the theoretical and practical aspects of genetics will be taught, including DNA/RNA structure, gene function, genome organization, inheritance, DNA replication and recombination and mutation. A major part of this course will cover various classical and modern molecular genetic techniques from the genomics era. The course includes individual and group exercises focusing on solving genetic problems. The knowledge gained in this course will show the student how to design and interpret genetic experiments, to answer relevant genetic questions. Students will need this knowledge for other courses including Molecular Biology, Laboratory Practical, Physical Biology of the Cell, and Evolutionary and Developmental Biology;	
Study Goals	At the end of this course, the student will be able to: 1. describe the function and organization of DNA in prokaryotes and eukaryotes (building blocks, genes, regulatory elements, gene layout) 2. formulate the basic principles of transcription and translation (genetic code, RNA and protein building blocks, splicing, non-coding RNA, RNA, layout, regulation of transcription and translation) 3. describe the function and organization of chromatin (histones, nucleosomes, chromosomes, centromers and telomers, eu- and heterochromatin, basic epigenetics) 4. understand and explain the concepts and terminology of classical genetics and inheritance (mendelian inheritance, dominance and recessiveness, homo- and heterozygote, X-linked, pedigrees, linkage, allele, population genetics) 5. describe how genomes are copied and transmitted (replication, mitosis, meiosis, recombination, linkage analysis) 6. understand and describe different types of mutations, their effects and chromosome instability (missense and nonsense mutations, deletion/insertion, LOH, translocation, CIN, transposons) 7. list causes of different genetic diseases and their therapies (gene therapy, genetic screening, hereditary syndromes, cancer, trinucleotide repeat disorder) 8. describe differences between the genetics of eukaryotes, prokaryotes and viruses (RNA/DNA viruses, bacteria) 9. understand both classical and modern molecular genetic techniques (genetic analysis, complementation analysis, mapping analysis, cytogenetics, forward/reverse genetic experiments, sequencing, PCR, expression analysis, microarrays, quantitative PCR, hiseq, CGH/tiling arrays, genomics) 10. interpret and critically evaluate results from genetic experiments 11. design genetic experiments to study the function of genes and their effect on phenotype, both individually as well as in groups (knockout, siRNA technology, recombinant DNA, cloning, expression and siRNA libraries)	
Education Method	Lectures Group Assignments (interpreting and designing genetic experiments)	
Literature and Study Materials	Genetics, a conceptual approach, 7th edition by Benjamin A. Pierce (Palgrave Macmillan/W.H. Freeman) And readers for assignments that will be distributed via Brightspace	
Assessment	1 test (30%) Final exam (60%) Presence and active participation during assignments (10%) In case of a re-exam, the grade for presence and active participation during assignments (10%) will remain and the resit of the exam will replace the grades of the test and the final exam and will count up to 90% of the final grade.	
Enrolment / Application	Enrollment in Brightspace is considered registration for assessment of non-written exam portion of the assessment, students must enroll in Brightspace for the course no later than the first week of classes. Registration for the written exam is via Osiris and deadlines are determined by TU Delft regulations on exam registration.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 may be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Remarks	Responsible Instructors: Prof. Dr. Ir. Jurgen Marteijn, j.marteijn@erasmusmc.nl, and Dr. Hannes Lans, w.lans@erasmusmc.nl, can be reached at their Erasmus MC email addresses.	
Studyload/Week	35 hours of lectures and group assignments for the total course of 4 ECTS. This indicates that approximately 80 hours of self-study is needed during the course. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Rotterdam	

NB1031	Introduction to studying Nanobiology	3
Responsible Instructor	Dr. J. Pothof	
Instructor	Prof.dr. R. Kanaar	
Contact Hours / Week x/x/x/x	9/0/0/0/0/0/0/0	
Education Period	1 1A	
Start Education	1 1A	
Exam Period	1 1A 1B	
Course Language	English	
Course Contents	In this course the students learn basic study skills and become familiar with the different forms of instruction and work (lectures, practical work, group work, problem sessions). Orientation sessions are aimed at providing group identity and cohesion for students in this new program. The students become familiar with team based learning, are divided into groups and practice this form of learning to complete a project on a topic in Nanobiology. The students will learn information processing skills; gathering information on a specific topic from the internet, evaluating the value of the information, summarizing information, ask questions and using the gathered information in a group project.	
Study Goals	At the completion of this course the student will be able to: 1. reflect on own academic and study skills and pitfalls 2. Work effectively in groups with respect for group members 3. Critically evaluate and provide positive feedback for working partners 4. Find basic scientific information on the internet 5. Understand the role of the different topics in the Nanobiology curriculum 6. Be familiar with, and adhere to, rules and regulations regarding academic integrity 7. Be able to ask meaningful questions	
Education Method	- Guided instruction and (short) lectures - Short writing assignments - Team based learning exercises - Group Assignments For 5 Week block scheduling this course will consist of sessions with mixed short lecture / instruction and (group) work assignments. Expect about 9 active working hours a week and equivalent out of class preparation time.	
Literature and Study Materials	Reading material to be handed out or posted on Brightspace	
Assessment	Grade will be pass / not pass There are 5 elements to evaluation and ALL have to be COMPLETED SATISFACTORY to pass. 1. Complete all short writing assignments (individual and group) 2. Peer evaluation of group work (individual) 3. Final project (report, poster)(group) 4. Attendance and active participation poster session and related writing assignment (individual) Resit: Students that failed item 1 - 3 will have the opportunity to redo the assignments. Item 4 (attendance and active participation in the poster session and related writing assignment) can only be done in person. Students that will not be able to participate or do not show up will have an opportunity next year (2025/26) to finish NB1031 by actively attending the poster session.	
Enrolment / Application	Registration for written exams are via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Remarks	Joris Pothof can be reached at j.pothof@erasmusmc.nl	
Studyload/Week	For 5 Week block scheduling this course will consist of 19 sessions of mixed short lecture / instruction and (group) work assignments. Expect about 9 active working hours a week and equivalent out of class preparation time. This is subject to change due to relevant Covid19 regulations. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Erasmus MC, Rotterdam	

NB1035	Programming Foundations	3
Responsible Instructor	Dr.ir. M.W. Docter	
Responsible Instructor	N. Šoštari	
Instructor	S. Dealy	
Contact Hours / Week x/x/x/x	0/0/0/0/12/0/0/0	
Education Period	3A	
Start Education	3A	
Exam Period	3A 3B	
Course Language	English	
Required for	Physics 1B, Biomolecular Programming, Bioinformatics, Computational Science, Electronic Instrumentation. Programming exercises also appear in several other Nanobiology courses.	
Expected prior knowledge	No prior knowledge is required.	
Course Contents	1. Theoretical background on computers and programming (algorithms and algorithmic thinking, libraries, types of programming languages, etc.). 2. Basic usage of command line, version control software, and an integrated development environment. 3. Troubleshooting/debugging, asking the right questions, and good coding practices. 4. Variables, data types, working with arrays (indexing, slicing, performing mathematical operations, etc.). 5. Flow control, Boolean logic to combine logical conditions, and loops. 6. Using functions from libraries and creating own functions. 7. Reading and writing data files, and plotting data.	
Study Goals	At the end of the course the student is able to: 1. Recognize basic concepts in working with computers and programming. 2. Use effectively programming software and supporting applications, in particular a computer terminal, an integrated development environment, and Git. 3. Use variables, functions, conditions, flow control, data files, and plotting to create Python code that addresses nanobiology problems. 4. Recognize and correct errors in Python code. 5. Apply style guidelines for code readability when writing Python code. 6. Review code created by others and provide useful feedback.	
Education Method	Sessions will include lectures, exercises, quizzes and graded assignments. Students will work on programming exercises, discuss code in groups and give peer feedback.	
Books	We will use an open online (Jupyter) book developed specifically for this course.	
Assessment	NB1035 is a practical course. For the evaluation, one of the two trajectories can be chosen at the beginning of the course. The choice made in the first week on the course will be considered a final choice and cannot be changed in later weeks. The exact deadline for making a choice will be communicated during the first lecture. If a student doesnt communicate a choice, as a default they will be evaluated according to trajectory I. ----------- Trajectory I "The training schedule": This trajectory helps with the planning of the course and allows the student to get credits for in-class quizzes and assignments. It is designed for those who have no prior experience or do not feel confident with programming. In trajectory I, the final grade will be composed of: - 90 % assignments - 10 % quizzes - mandatory peer feedback There will be 3 peer feedback moments during the course, scheduled in the same weeks as graded assignments. Participation in peer feedback will be a prerequisite for taking part in the assignment of the corresponding week. Peer feedback will not be graded (i.e., formative assessment). There will be 3 graded assignments solved in class, each counting for 30 % of the final grade. For a passing grade, it is mandatory to take part in all 3 assignments and achieve a score of at least 5 at each of the assignments. There will be 5 quizzes, the best 4 (with a grade >=0) will each count for 2.5% of the final grade. The quizzes will be offered in class. Participation in quizzes is not mandatory, i.e., the student can also get a passing grade of max. 9 through graded assignments only. ----------- Trajectory II "Experienced programmer": This trajectory is designed for those who feel confident with programming. Taking part in the peer feedback sessions and quizzes is still recommended but is not obligatory. We will provide a pre-test on the first day of the course, such that students can make an informed choice about taking trajectory II. In trajectory II, the final grade will be composed of: - 100 % assignments There will be 3 graded assignments solved in class, each counting for 33.3 % of the final grade. For a passing grade, it is mandatory to take part in all 3 assignments and achieve a score of at least 5 at each of the assignments. ----------- The schedule of the quizzes and assignments is posted in Brightspace no later than the first day of class. You will have 2 opportunities to redo an assignment, once in week 5 and once in week 10. - You may only attempt each assignment twice in an academic year. - You will not be able to redo all three assignments. - Missing an assignment during class will limit your opportunities for redoing an assignment, since you will need to do the one you missed during the make-up sessions. - You must indicate when requesting to participate in the make up or resit session which assignment you wish to do. - The best grade will be used in final calculations. *Transition rules* NB1035 will only be offered in 2024/2025. Students who fail it will need to take the new programming course NB1420 Programming Foundations which will be in Q2 and 5 EC.	

Permitted Materials during Tests	Graded assignments must be solved individually and with restricted resources, i.e., no communication with other people, no internet, and no large language model usage allowed. Assignments must be solved in class and submitted during the class, in the scheduled time. A reference sheet will be provided to students during the assignment.
Enrolment / Application	To join the course, students need to enroll in BrightSpace in the first week of the course. Enrolling in Brightspace also means enrolling for the assessments during the course. Students must sign up for make up/resit sessions via Brightspace at least 3 business days before the session takes place.
Studyload/Week	Aside from 12 contact hours per week, the students are expected to spend additional 9 hours studying for this course (mainly continue the exercises started on campus and studying with the course book). Note that this is only an estimate and not a guarantee, as different students learn at different paces.
Location	Delft and Rotterdam

NB1052	**Journal Club 1**	**3**
Responsible Instructor	G. Roshchupkin	
Responsible Instructor	D. Andrade De Jesus	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In Journal Club several scientific research skills will be taught and practiced. Using examples from scientific literature and popular press the students become acquainted with current research in biomedicine, nanoscience, biophysics and related fields. Introductory sessions will focus on how to find scientific information in print and electronic form, the various types of information sources, how to evaluate the worth of different information sources and specificially how to read a scientific research article. Working sessions will be team based learning where each student group will be given an article to read and a set of questions to answer based on the article. Finally student groups will give a presentation on the article they have read and relevant background articles, based on assigned questions. This course utilizes and builds on team learning skills introduced in Intro to Nanobiology. The skills of processing scientific and research information are required for subsequent courses.	
Study Goals	At the completion of this course the student will be able to: 1. identify different information sources commonly used in scientific research and describe their characteristics and use. 2. identify the type of information found in the different parts of a research article (abstract, introduction, materials and methods, results, discussion) 3. use internet resources to search for and retrieve research articles, and related background information. 4. work effectively as a team to answer a set of specific questions and create a presentation 5. constructively evaluate peer group members with respect to contribution to group assignments 6. identify alternative explanations for aspects of published research as a step toward critical evaluation 7. discuss the difference between scientific information presented in research articles and in the popular press.	
Education Method	- Short introduction and instruction lectures - Meetings of groups to work on assignments, team-based learning - Group presentation This course will consist of sessions including short instructions and group work activities, with variable amounts of additional preparation and reading required. NOTE: we will monitor current policy developments and regulations to decide whether in-person gatherings for groupwork and on-campus instruction will be possible.	
Assessment	This is a graded course. Final grade is determined as follows: 90% final group presentation 10% based on peer feedback on contributions to group work Further, students have to pass an online information literacy course.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Studyload/Week	Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Rotterdam	

NB1102	Chemistry 1	3
Responsible Instructor	Dr. B.C. Tilly	
Instructor	Dr. J.A.A. Demmers	
Instructor	Dr. J.A. van der Knaap	
Contact Hours / Week x/x/x/x	0/6/0/0/0/0/0/0	
Education Period	1 1B	
Start Education	1B	
Exam Period	1 2 1B 2A	
Course Language	English	
Course Contents	Chemistry studies the composition, properties and transformations of matter. Chemical principles apply to living organisms and knowledge of chemistry is essential for the understanding of biological processes. Understanding chemistry is especially important for research at the nanoscale, where complex interactions between different molecules are governed by chemical forces and barriers. In this course the student will learn the fundamental properties of atoms and molecules, how they react with each other and obey the laws of thermodynamics. In addition the student will obtain basic knowledge of analytical and spectroscopic techniques used in life sciences. The student is expected to be familiar with the concepts and skills of Math as taught during the course Analysis 1. The knowledge obtained during this course will be applied in other biologically orientated courses including Biochemistry, Biophysics, Evolutionary & Developmental Biology, Lab Practicum and Physical Biology of the Cell.	
Study Goals	At the completion of this course the student should be able to: 1. Understand the structure and properties of atoms in terms of quantization, wavefunctions, atomic orbitals and periodicity. 2. Describe the characteristics of the Lewis, valence bond, and molecular orbital theories of chemical bonds 3. Describe the characteristics of ionic, covalent and metallic bonding and predict the type of bonding from the electronegativities of the atoms 4. Recognize non-covalent interactions and predict physical properties of atoms and molecules. 5. Predict the shape of molecules using valence shell electron pair repulsion theory (VSEPR) 6. Construct molecular orbital energy level diagrams for diatomics 7. Recognize functional groups in organic molecules and name these compounds using IUPAC nomenclature and common names. 8. Know the Arrhenius, Brønsted-Lowry and Lewis definitions of acids and bases and calculate the pH of solutions of acids, bases and buffers 9. Describe and predict the behaviour of gases using the ideal gas law.	
Education Method	This course contains lectures, tutorials and home assignments.	
Literature and Study Materials	Chemistry3. Fourth edition. Burrows, Holman, Parsons, Pilling and Price, Oxford University Press, 2021. Achieve Macmillan learning (subscription required, provided by Nanobiology programme for nanobiology students).	
Assessment	Final grade for the course is based on the exam plus 0.1 point bonus for each sufficiently completed home assignment (max. 0.7 points). The bonus will be received only once and will be added to your grade the first time the exam is taken (maximum grade is 10).	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Remarks	Ben Tilly can be reached at b.tilly@erasmusmc.nl	
Studyload/Week	The course (5 weeks) consists of lectures, online and in class tutorials (35h), home assignments (11h) and 38h of study at home and exam preparation. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Rotterdam	

NB1110	Chemistry 2	3

Responsible Instructor	Dr. B.C. Tilly
Instructor	Dr. J.A.A. Demmers
Instructor	Dr. J.A. van der Knaap
Contact Hours / Week x/x/x/x	0/0/8/0/0/0/0/0
Education Period	2 2A
Start Education	2
Exam Period	2A 2B
Course Language	English
Course Contents	Chemistry studies the composition, properties and transformations of matter. Chemical principles apply to living organisms and knowledge of chemistry is essential for the understanding of biological processes. Understanding chemistry is especially important for research at the nanoscale, where complex interactions between different molecules are governed by chemical forces and barriers. In this course the student will learn the fundamental properties of atoms and molecules, how they react with each other and obey the laws of thermodynamics. In addition the student will obtain basic knowledge of analytical and spectroscopic techniques used in life sciences. The student is expected to be familiar with the concepts and skills of Math as taught during the course Analysis 1. The knowledge obtained during this course will be applied in other biologically orientated courses including Biochemistry, Biophysics, Evolutionary & Developmental Biology, Laboratory Practicum, and Physical Biology of the Cell.
Study Goals	At the completion of this course the student will should be able to: 1. Describe and explain the First Law of Thermodynamics and calculate the enthalpy and internal energy changes for a chemical reaction 2. Describe and explain the Second Law of Thermodynamics and calculate the entropy and Gibbs energy changes of a reaction 3. Define and use the thermodynamic equilibrium constant and know how it depends on reaction conditions 4. Know what is meant by the rate of the reaction and how it depends on reactant concentrations, temperature and the presence of catalysts, including enzymes 5. Recognize structural isomers and stereoisomers 6. Understand the basic principles of molecular spectroscopy in terms of quantization of molecular energy and transitions between molecular energy levels when matter interacts with radiation 7. Describe electrochemical, chromatographic and spectroscopic methods of analysis and suggest suitable methods for a given analyte. 8. Determine the structure of an unknown compound by using mass spectra, IR spectra and NMR spectra
Education Method	This course contains lectures, tutorials and home assignments.
Literature and Study Materials	Chemistry3. Fourth edition. Burrows, Holman, Parsons, Pilling and Price, Oxford University Press, 2021. Achieve Macmillan learning (subscription required).
Assessment	Final grade for the course is based on the exam plus 0.1 point for each sufficiently completed home assignment (max. 0.7 points). The bonus will be received only once and will be added to your grade the first time the exam is taken (maximum grade is 10).
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.
Remarks	Ben Tilly can be reached at b.tilly@erasmusmc.nl
Studyload/Week	The course (5 weeks) consists of lectures, online and in class tutorials (35h), home assignments (11h) and 38h of study at home and exam preparation. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	Rotterdam

NB1120	Biomolecular Programming	3
Responsible Instructor	Prof.dr. A.B. Houtsmuller	
Responsible Instructor	Dr. J.A. Slotman	
Responsible Instructor	Dr.ir. G.J. Kremers	
Instructor	Dr. J.A. Slotman	
Instructor	Dr.ir. G.J. Kremers	
Contact Hours / Week x/x/x/x	0/0/0/0/0/9/0/0	
Education Period	3 3B	
Start Education	3 3B	
Exam Period	3 4 3B 4A	
Course Language	English	
Course Contents	In the course Biomolecular Programming (BMP), students learn how to design algorithms for performing specific tasks and to implement algorithms in simple and advanced computer programs in the programming language PYTHON. They learn how to use the acquired programming skills to design and implement Computer simulation programs to simulate the behaviour of biomolecules in the cell and the cell nucleus. BMP is complementary to various topics in molecular biology and related courses since the insight in cellular dynamics obtained by simulating and visualising dynamic behaviour of biomolecules provides a strong basis to understand how biophysical and biochemical principles apply to the function of the living cell, or living systems. BMP links to the Analysis and Differential Equations courses, where skills are taught required to establish analytical models to describe the modeled systems. BMP also links to Physical Biology of the Cell since the simulation software package BSL, is largely based on algorithms implemented by the students during the course.	
Study Goals	At the completion of this course the student will be able to: 1. Demonstrate understanding of the concept ALGORITHM (in the strict meaning) 2. Design flow diagrams to schematically describe the subsequent steps and loops within an ALGORITHM 3. Demonstrate the capacity to translate (=implement) an ALGORITHM into a (part of a) COMPUTER PROGRAM, including the specific skills to use in PYTHON: - Variables - IF-THEN-ELSE statements and the Boolean operators AND, OR, XOR, and NOT - FOR and WHILE loops and the integer operators MOD and DIV - ARRAY Variables 4. Demonstrate the ability to combine the above skills to implement algorithms to algorithms to simulate movement and diffusion of biomolecules within and between compartments and the interaction of individual molecules with each other (binding, modification, degradation). 5. Demonstrate understanding of biomolecular dynamic equilibria and steady state systems	
Education Method	- Computer Practicals including introductory short lectures (20-30 min.) - Individual assignments on computer programming	
Literature and Study Materials	-Instruction Binder Python Computer Programming, description and explanation of the basic programming elements and exercises (The binders are developed by the instructor team. They are free and will be distributed at the start of the course) -Software - Pycharm: a PYTHON programming environment (The software is free and will be distributed at the start of the course) -Laptop (all operating systems are allowed) (Each student should always bring his/her own laptop)	
Assessment	The assessment consist of: Making a working final exercise and discussing this program (15 min) The resit consists of: Making a working final exercise and discussing this program (15 min) The final grade is composed as follows: Pass or Fail Registration for exams is done via: Brightspace. One day before the scheduled exam the code of your final exercise should be handed in via brightspace.	
Enrolment / Application	Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The teachers can be reached as A.B. Houtsmuller (a.houtsmuller@erasmusmc.nl) Johan Slotman <j.slotman@erasmusmc.nl> Gert-Jan Kremers <g.kremers@erasmusmc.nl> The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Location	Erasmus MC / TUDelft	

NB1132	Biophysics	3
Responsible Instructor	Prof.dr. C. Joo	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/10/0	
Education Period	4A	
Start Education	4A	
Exam Period	4A 4B	
Course Language	English	
Course Contents	In recent decades, physicists have developed advanced tools that enable us to observe the inner workings of cells in greater detail than ever before. This wealth of new biological information has led to exciting new areas of research in understanding biological processes. This course aims to teach students the fundamental physical concepts that form the basis of these biological phenomena. Topics covered include diffusion, biopolymers, motor proteins, and stochasticity. Students will build on concepts previously introduced in courses such as Physics 1, Biochemistry, Molecular Biology, and Analysis. In addition, students will learn how to use programming to model diffusion and biopolymers, using previous introduction to Python. Overall, this course offers a unique opportunity to gain a deeper understanding of the physical principles behind the complex processes that occur within living cells.	
Study Goals	By the end of this course, students will have gained the following skills: 1. The ability to describe the basic length and time scales that are relevant to biological processes at the molecular and cellular levels. 2. The ability to quantitatively describe the mechanical properties of biopolymers. 3. An understanding of the laws of motion at low Reynolds numbers. 4. The ability to model diffusive motion in one to three dimensions using both theoretical concepts and simulation techniques. 5. An understanding of how to apply the Boltzmann distribution to describe biological processes. Overall, this course will equip students with a broad range of skills and knowledge that will enable them to better understand and analyze biological processes at the molecular and cellular levels.	
Education Method	- Weekly lectures - Weekly problem sets and discussion - Weekly programming sessions	
Books	Physical Biology of the Cell (Phillips, Kondev, Theriot)	
Assessment	Final report: 10% Final exam: 90% -- Resit Final report: 10% (the grade from above carries forward) Retake exam: 90%	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Studyload/Week	This course is worth 3 ECTS, which corresponds to approximately 84 hours of work. Over the course of four weeks, students can expect to spend around 10 hours per week attending lectures and exercise classes, totaling approximately 37 hours. This leaves approximately 8 hours per week for independent study and review at home, as well as an additional 15 hours for exam preparation. It is important to note that every student is different and may require more or less time to master different topics. This is a rough estimate and should not be taken as a guarantee of how much work each individual student will need to put in to succeed in the course.	
Location	Delft	

NB1140	Physics 1a	4
Responsible Instructor	Dr. J.W. Zwanikken	
Contact Hours / Week x/x/x/x	0/10/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Physics 1B, Physics 2, Biophysics, Thermodynamics and Transport, Statistical Physics, Physical Biology of the Cell	
Expected prior knowledge	Analysis 1	
Course Contents	The courses Physics 1A and 1B introduce the basic principles of classical physics. Physics 1A covers classical mechanics and an introduction to continuum physics. All the physical principles will be taught with biological examples.	
Study Goals	At the completion of this course the student will be able to: 1. Explain the basic laws of classical mechanics, applied to both particles and continuous materials. 2. Derive and solve equations of motion for common situations in classical physics, including conservative forces, oscillators and ideal fluid flow. 3. Apply the laws and equations listed above to various real-life situations. 4. Analyze macroscopic and microscopic biological phenomena quantitatively. 5. Apply the basic physical concepts learned here to advanced biology and physics topics (Thermodynamics and Transport; Statistical Physics) and biophysics topics (Biophysics; Physical Biology of the Cell; and Nanotechnology).	
Education Method	Live lectures, demonstrations, and work sessions, that cover the entire course, and online materials that also cover the entire course. A choice can be made to follow the live sessions, or to study the online materials, or to use a combination of both. For the evaluation, two trajectories can be chosen at the beginning of the course: Trajectory I "The training schedule": - 6 preparatory quizzes count for 12% of the final grade - 8 homework sets count for 12% of the final grade - the midterm exam counts for 36% of the final grade - the final exam counts for 40% of the final grade This trajectory helps with the planning of the course, may save the participant a lot of energy figuring out how to prepare for the exam, and allows the participant to get credits for quizzes and homework. It is designed particularly for those who do not feel so confident with physics, or with planning or managing a larger course. Trajectory II "The independent" - the midterm exam counts for 47% of the final grade - the final exam counts for 53% of the final grade This trajectory allows the participants to follow their own schedule, to take part on the quizzes and work sessions, but does not award credits for the quizzes and homework sets. This trajectory is designed for those who feel more confident with physics, those who have the discipline (or energy) to manage their own course, or those who want to practice these important academic skills. Taking part on the work sessions and quizzes is still strongly recommended, but is not obligatory. The choice of trajectory needs to be communicated via Brightspace during the first week of the course, and cannot be changed during the quarter.	
Literature and Study Materials	Wolfson, Essential University Physics, 3rd edition, including Mastering Physics. NB: You need the FULL version of the book, NOT the 'special TU Delft version' (which is the same book minus the chapters on thermodynamics). The book comes in a 2-volume package and is also used in physics 1B and physics 2. So please buy both volumes 1 and 2, and the Mastering Physics. In addition to the book, lecture notes and slides will be provided online.	
Assessment	Assessment: Trajectory I "The training schedule": - 6 preparatory quizzes count for 12% of the final grade - 8 homework sets count for 12% of the final grade - the midterm exam counts for 36% of the final grade - the final exam counts for 40% of the final grade Trajectory II "The independent" - the midterm exam counts for 47% of the final grade - the final exam counts for 53% of the final grade The midterm exam covers the first half of the course, and the final exam covers the entire course, with a strong emphasis on the topics of the second half. The retake exam covers the entire course, in proportion to the attention that the topics received during the quarter.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Studyload/Week	Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Continuing Courses	Physics 1B and Physics 2.	

NB1143	Physics 1B	3
Responsible Instructor	Dr. J.W. Zwanikken	
Instructor	Dr. S.M. Depken	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/0/9	
Education Period	4B	
Start Education	4B	
Exam Period	4B 5AB	
Course Language	English	
Course Contents	In Physics 1B we continue the exploration of classical physics from Physics 1A, and provide an introduction to electromagnetism. The course consists of a theoretical and an exercise part. The exercise part covers masteringphysics and electric circuit simulations, as well as homework assignments that will be graded using peer grading. Practically the course consists of plenary lectures followed by working sessions in three subgroups supervised by a TA.	
Study Goals	At the end of this course you will be able to (Learning Objectives, LO): 1. Explain and use the basic laws of electrostatics to calculate electric forces, fields and potentials. 2. Explain and use the laws of magnetism to calculate magnetic forces, fields. 3. Apply the laws of electromagnetism to explain and analyze the action of standard circuit elements (resistor, capacitor, and inductor). 4. Analyze standard element alternating-current circuits in terms of frequency response and resonances. 5. Write clear and detailed solutions to text book problems, such that they can be understood by your peers.	
Education Method	Lectures and exercises Activities: Mastering physics: LO1 and LO2 Homework: LO1, LO2, LO3, LO4 Peer assessment of Homework: LO5 Simulations: LO1, LO2, LO3	
Literature and Study Materials	Required: - Essential University Physics, Richard Wolfson, Addison-Wesley - Mastering Physics login code, Richard Wolfson, Addison-Wesley The book comes in a 2-volume package that is also used in Physics 1A and Physics 2. So please buy both volumes 1 and 2, and the Mastering Physics. If you buy the package second-hand, check that the Mastering Physics license is still valid! (It typically is valid for two years).	
Assessment	To pass you always need to score at least 5.8 on the final exam, and have a passing grade from mastering physics. The final grade will then be calculated as: Homework: If the best 3 out of 4 HW sets increase the overall grade, they will account for 20% of final grade. Out of this, 15% will be determined by grade on HW solutions handed in, and 5% on the grade for the peer grading performed. Final Exam: 80% (if HW counted)/(100% if HW not counted) Masteringphysics: pass/fail @Retakers If you passed masteringphysics in a previous year, you do not need to re-do this, though your old homework will not count towards the final grade.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and likely be larger and may be at a different time of year than the old course.	
Studyload/Week	Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Delft	

NB1150	Lab Course Track A-1	3
Responsible Instructor	Dr. L. Mezzanotte	
Responsible Instructor	A.F. Theil	
Instructor	Dr. J. Pothof	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This practical course is the first experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, biochemistry, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used bio-analytical methods, process and evaluate data and concisely report their findings.	
Study Goals	At the end of the course the student will be able to: 1. Understand the safety rules in laboratories and know how to follow them; 2. Maintain a clean and organised lab bench; 3. Keep a digital lab journal in an organised fashion and write lab reports clearly; 4. Understand the ethical importance of keeping an accurate and complete lab journal; 5. Master pipetting; 6. Isolate plasmid DNA and perform digestion reaction with restriction enzymes; 7. Perform PCR Gel electrophoresis; 8. Design primers and understand the basics of recombinant DNA technology for cloning; 9. Apply different techniques to determine protein concentration; 10. Purify a protein using Ion-exchange chromatography; 11. Apply a method to measure enzymatic activity; 12. Ion-Exchange Chromatography; 13. Apply tests of significance to datasets.	
Education Method	This course consists of a series of 2-hour theoretical lectures, introducing the theory underlying the practical work and providing deepening/broadening background information. Followed by a 4-hour supervised session of practical laboratory work. Additionally, data analysis will take approximately 2 hours in class or at home, as well as time for homework and lab journal maintenance.	
Course Relations	There are four Nanobiology 1st year lab courses: NB1150 Lab Practicum A1 NB1151 Lab Practicum A2 NB1163 Lab Practicum B1 NB1164 Lab Practicum B2 Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.	
Practical Guide	Completion of the lab journal of the previous week is mandatory to get access to the practical of the following week. Students with an incomplete lab journal will be denied access to the following practical.	
Assessment	The final grade is based on a written exam and report on your experiments. In addition, we will monitor the following three items with a fail-pass: 1. The quality of your lab journal and its completion before every practical 2. 100% attendance during the practicals 3. 2 out of 2 assignments should be performed correctly and handed in timely A single fail will result in us not evaluating your exam. In that case, you will need to update your lab journal and/or redo (a) practical(s).	
Enrolment / Application	This class is only for students in the Nanobiology BSc programme. Students who wish to take the course will be assigned into this course or its partner course, NB1163. Students who take this course, must take NB1151 as their second lab course to fulfill this requirement. Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing info-bsc-nb@tudelft.nl. Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. Because this is a practical course resits will be limited in 2025-2026 and transition rules for students who don't complete this course will apply. The new courses will be larger.	
Studyload/Week	3ECTS = 84 hours = 8.4 hours/week Lectures and practicals 6 hrs/week * 7 weeks = 42 hours Homework 2 hrs/week * 8 weeks = 16 hours Exams (midterm and final) 3 hrs/week * 2 weeks = 6 hours Study course material at home 2hrs/week * 10 weeks = 20 hours Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Erasmus MC	

NB1151	Lab Course Track A-2	3

Responsible Instructor	Dr. H.J. Geertsema
Instructor	Dr. M.E. Aubin-Tam
Contact Hours / Week x/x/x/x	0/0/0/5
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	English
Course Contents	This practical course is the first experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used microscopy methods, process and evaluate data and concisely report their findings.
Study Goals	At completion of this course, you will be able to: 1. Identify and discuss measures to improve the environmental impact of your own experiments 2. Employ a light microscope to visualize and obtain quantitative information on a self-prepared sample and on microscope performance 3. Employ Atomic Force Microscopy to study and obtain quantitative information on a self-prepared sample and AFM performance 4. Analyze the data obtained from your experiments and clearly and concisely report on the findings, including making graphical representations 5. Develop and execute an experiment to probe a physical property of your sample of choice 6. Concisely and clearly report in a digital lab journal
Education Method	This course consists of a theoretical lecture, introducing the theory underlying the practical work and providing deepening/broadening background information, a supervised practical session and data analysis. Some homework will be required.
Course Relations	There are four Nanobiology 1st year lab courses: NB1150 Lab Practicum A1 NB1151 Lab Practicum A2 NB1163 Lab Practicum B1 NB1164 Lab Practicum B2 Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.
Assessment	The course grade is calculated by a weighted average of the Exam grade (65%) and the individual assignment (35% total, 5% per assignment). The final exam covers all the experimental skills obtained during the course. The individual assignments are to be handed in on time and are graded from 1-10. Deliverable dates for the assignments are communicated through the electronic learning environment. In addition, we will monitor the following two items with a fail-pass: 1. The quality of your lab journal 2. 100% attendance during the practical sessions Notes: - No scores and/or submissions are transferred to the next year. - In order to pass the course, students need to score at least a 5.75 during the final exam - Attendance to all practical sessions is mandatory, exceptions are possible based on a written request with justification (serious illness, family emergencies) to the responsible teacher, Dr. Hylkje Geertsema (h.j.geertsema@tudelft.nl).
Enrolment / Application	Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing info-bsc-nb@tudelft.nl. Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. Because this is a practical course resits will be limited in 2025-2026 and transition rules for students who don't complete this course will apply. The new courses will be larger.
Studyload/Week	3 ECTS = 84 hours = 8.4 hours/week We expect you to spend on: - Lectures: 2 hours/week * 10 weeks = 20 hours - Lab work: 4 hours/week * 8 weeks = 32 hours - Homework: 3 hours/week * 10 weeks = 30 hours - Exam prep:2 hours/week * 1 week = 2 hours Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.

NB1163	Lab Course Track B-1	3
Responsible Instructor	Dr. H.J. Geertsema	
Instructor	Dr. M.E. Aubin-Tam	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This practical course is the first experimental experience the students of the Nanobiology degree will receive. The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used microscopy methods, process and evaluate data and concisely report their findings.	
Study Goals	At completion of the course, you will be able to: 1. Conduct basic lab technical skills while applying safety rules(i.e. pipetting, preparing buffers, executing a lab protocol) 2. Identify and discuss measures to improve the environmental impact of your own experiments 3. Employ a light microscope to visualize and obtain quantitative information on a self-prepared sample and on microscope performance 4. Employ Atomic Force Microscopy to study and obtain quantitative information on a self-prepared sample and AFM performance 5. Analyze the data obtained from your experiments and clearly and concisely report on the findings, including making graphical representations 6. Develop and execute an experiment to probe a physical property of your sample of choice 7. Concisely and clearly report in a digital lab journal	
Education Method	This course consists of a theoretical lecture, introducing the theory underlying the practical work and providing deepening/broadening background information, a supervised practical session and data analysis. Some homework will be required.	
Course Relations	There are four Nanobiology 1st year lab courses: NB1150 Lab Practicum A1 NB1151 Lab Practicum A2 NB1163 Lab Practicum B1 NB1164 Lab Practicum B2 Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.	
Assessment	The course grade is calculated by a weighted average of the Exam grade (65%) and the individual assignment (35%, 5% per assignment). The final exam covers all the experimental skills obtained during the course. The individual assignments are to be handed in on time and are graded from 1-10. Deliverable dates for the assignments are communicated through the electronic learning environment. In addition, we will monitor the following two items with a fail-pass: 1. The quality of your lab journal 2. 100% attendance during the practical sessions Notes: - No scores and/or submissions are transferred to the next year. - In order to pass the course, students need to score at least a 5.75 during the final exam - Attendance to all practical sessions is mandatory, exceptions are possible based on a written request with justification (serious illness, family emergencies) to the responsible teacher, Dr. Hylkje Geertsema (h.j.geertsema@tudelft.nl)	
Enrolment / Application	This class is only for students in the Nanobiology BSc programme. Students who wish to take the course will be assigned into this course or its partner course, NB1150. Students who take this course, must take NB1164 as their second lab course to fulfill this requirement. Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing info-bsc-nb@tudelft.nl. Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. Because this is a practical course resits will be limited in 2025-2026 and transition rules for students who don't complete this course will apply. The new courses will be larger.	
Studyload/Week	3 ECTS = 84 hours = 8.4 hours/week We expect you to spend on: - Lectures: 2 hours/week * 10 weeks = 20 hours - Lab work: 4 hours/week * 8 weeks = 32 hours - Homework: 3 hours/week * 10 weeks = 30 hours - Exam prep:2 hours/week * 1 week = 2 hours Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	TUDelft	

NB1164	Lab Course Track B-2	3

Responsible Instructor	Dr. L. Mezzanotte
Responsible Instructor	A.F. Theil
Instructor	Dr. J. Pothof
Contact Hours / Week x/x/x/x	0/0/0/5
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	English
Course Contents	The main aim of this course is to introduce the students to an experimental environment in which they will learn the essential skills to perform experiments in the area of nanobiology, biochemistry, physics and chemistry. In particular, students will learn how to setup a basic scientific inquiry, keep track of the experimental approach, handle commonly used analytical methods, process and evaluate data and concisely report their findings.
Study Goals	At the end of the course the student will be able to: 1. Keep a digital lab journal in an organised fashion and write lab reports clearly; 2. Understand the ethical importance of keeping an accurate and complete lab journal; 3. Master pipetting; 4. Isolate plasmid DNA and perform digestion reaction with restriction enzymes; 5. Perform PCR Gel electrophoresis; 6. Design primers and understand the basics of recombinant DNA technology for cloning; 7. Know the basics of bacterial culture and apply optical detection methods to image luminescent bacteria; 8. Apply different techniques to determine protein concentration; 9. Purify a protein using Ion-exchange chromatography; 10. Apply a method to measure enzymatic activity; 11. Ion-Exchange Chromatography; 12. Apply tests of significance to datasets.
Education Method	This course consists of a series of 2-hour theoretical lecture, introducing the theory underlying the practical work and providing deepening/broadening background information. Followed by a 4-hour supervised session of practical laboratory work. Additionally, data analysis will take approximately 2 hours in class or at home, as well as time for homework and lab journal maintenance.
Course Relations	There are four Nanobiology 1st year lab courses: NB1150 Lab Practicum A1 NB1151 Lab Practicum A2 NB1163 Lab Practicum B1 NB1164 Lab Practicum B2 Students take either A1 and A2, or B1 and B2. Other combinations are not permitted. Students are assigned into group A or group B by the programme based on course capacity. If students from previous years need one more lab course, they need to discuss with the Study Adviseur which course to take.
Practical Guide	Student must complete the lab journal of previous week to access to practical of the week after. Students who do not complete the weekly lab journal assignment will be denied access to the following practical.
Prerequisites	Students must have successfully completed NB1163 and passed lab journal before completing this course.
Assessment	The final grade is based on a written exam and report on your experiments. In addition, we will monitor the following three items with a fail-pass: 1. The quality of your lab journal and its timely completion 2. 100% attendance during the practicals 3. 2 out of 2 assignments should be performed correctly and handed in timely A single fail will result in us not evaluating your exam. In that case, you will need to update your lab journal and/or redo (a) practical(s).
Enrolment / Application	Exchange or bridging students wishing to take a Nanobiology lab course must contact the Nanobiology programme coordinator two weeks before the course starts by emailing info-bsc-nb@tudelft.nl. Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. Because this is a practical course resits will be limited in 2025-2026 and transition rules for students who don't complete this course will apply. The new courses will be larger.
Studyload/Week	3ECTS = 84 hours = 8.4 hours/week Lectures and practicals 6 hrs/week * 7 weeks = 42 hours Homework 2 hrs/week * 8 weeks = 16 hours Exams (midterm and final) 3 hrs/week * 2 weeks = 6 hours Study course material at home 2hrs/week * 10 weeks = 20 hours Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.

NB1201	Analysis 1	5
Responsible Instructor	Dr. F.J. van de Bult	
Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2A	
Course Language	English	
Required for	This course will teach the basic mathematics which will be relevant to many other courses in the Nanobiology programme. Some examples of courses which rely especially on this course are Analysis 2 and 3 (NB1206 and NB1211) Physics 1A, 1B and 2 (NB1140, NB1143, NB2141) Differential Equations (NB2191)	
Expected prior knowledge	High school mathematics, to be specific the contents of the Dutch Mathematics B. That is: * Functions: Polynomials, Power functions, Trigonometric functions, Exponential functions and logarithms. * Algebraic manipulation of formulas * Standard graphs of functions * Differentiation (including product rule, chain rule, quotient rule). * Integration basics	
Course Contents	In this course you continue the study of functions f(x) of one variable that you started in high school. You will learn a few more techniques for evaluating integrals. You will learn how to find functions that satisfy differential equations such as F = ma (which involves the second derivative a of the location). To do this, it turns out we also have to consider solutions to x^2 = -1 (and yes, this equation will have solutions.) You will also study what happens when you add infinitely many (tiny) numbers. One important application of such a sum is that it can be used to find better and better approximations of quantities and functions which cannot be calculated explicitly. Topics of Analysis: complex numbers, linearization, methods of integration, ordinary differential equations, (Taylor) series.	
Study Goals	After this course the student should have achieved the following learning goals: 1. The student can apply differentiation in various settings: a. Using (inverse) trigonometric and hyperbolic functions, b. For implicitly defined functions, c. To approximate functions and changes in function values d. In applications of the Fundamental Theorem of Calculus 2. The student can apply integration of univariate functions in various settings: a. Calculate integrals using the substitution rule, partial fractions, and integration by parts. b. Determine convergence (and value) of improper integrals 3. The student can work with univariate ordinary differential equations: a. Approximate solutions using Euler's rule b. Solve 1st order linear and separable equations c. Solve 2nd order linear equations with constant coefficients. 4. The student can work with complex numbers a. Do basic calculations b. Work with polar form of complex numbers c. Solve simple equations involving complex numbers (quadratic equations and binomial equations) 5. The student can work with series a. Apply tests to determine convergence of a series ("divergence", integral, comparison, ratio, and alternating series tests) b. Obtain error bounds on the difference of a full series and partial sums 6. The student can work with power series a. Determine interval and radius of convergence. b. Determine Taylor series of a function c. Work with Taylor's theorem on the error of a Taylor polynomial 7. For all the above topics the students can give a few situations (esp. outside mathematics) where it can be applied and is able to apply them to those situations.	
Education Method	(Assuming no new COVID measures) This course will include the following educational activities: - Interactive lectures - Work sessions where you can practice under the guidance of student assistants - Computer graded homework to do at home or in the work sessions. Apart from basic questions where you can directly apply the methods learned in class, the homework will also contain questions where you have to be more creative. This will help develop your problem solving skills. (The exam can contain such a more creative question or require one of the techniques you had to discover for yourself in one of the homework problems.)	
Computer Use	Digitally corrected exercises are part of the homework. A digital book is used for the course.	
Literature and Study Materials	- Openstax Calculus volumes 1, 2, and 3. Download the original books for free from openstax.org, or access them online through Brightspace. - There will be some lecture extra notes on complex numbers written by the lecturer The lecture notes are available on Brightspace.	
Assessment	(Assuming no new COVID measures) The (regular) assessment consists of - Two written exams (2x1.5 hours) (Grades M and F) - There are quizzes, which count for 20%. For each week of lectures there is a quiz in the digital exercise system which you can take multiple times. The best 6 out of 8 quiz results count. If you actively participate in at least 6 quizzes, the quiz-grade is taken to be a minimum of 5.75 (so you can't fail due to bad quiz results). Grade Q.	

	The final grade of regular part is thus 2/5 * M + 2/5 * F + 1/5 * Q Students who take this course for a second year do not have to do all the quizzes. In that case their grade is 1/2 * M + 1/2 * F. The resit is a single 3 hour exam, for which the quizzes do not count. You have to register in Osiris for both exams and the resit. You do not have to register for the quizzes. Intermediate grades for the regular exam (M, F or Q) cannot be kept for future years. The compensation rule Analysis for Nanobiology is applicable. If the grades for Analysis 1, 2, and 3 (NB1201, NB1206, and NB1211 or prior incarnations) are all at least 5.0 (in decimals), and the weighted (by EC) average of these grades equals at least 5.8 after rounding to decimals, then the student gets the EC for all three courses.
Permitted Materials during Tests	Pen, pencil and a simple, non-graphing calculator. The calculator is not allowed to be able to do calculations with complex numbers or evaluate integrals (either analytically or numerically). As such the Casio fx-570, fx-580, fx-991 and TI-36x are not allowed. Allowed are, for example, Casio fx-82 or TI-30.
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments (such as quizzes), students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and and may be at a different time of year than the old course.
Studyload/Week	5ECTS = 140 hours total Regular weeks (8 times): 6 hour lecture + 4 hour work session + 4 hour self study = 14 hour/week Exam week (2 times): 12.5 hour exam preparation + 1.5 hour exam = 14 hour/week Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	TU Delft

NB1206	Analysis 2	5
Responsible Instructor	Dr. F.J. van de Bult	
Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/10/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3A	
Course Language	English	
Required for	This course will teach some of the basic mathematics which will be relevant to many other courses in the Nanobiology programme. Some examples of courses which rely especially much on this course are Analysis 3 (NB1211) Physics 1A, 1B, 2 (NB1140, NB1143, NB2141) Signals and Systems (TN2545)	
Expected prior knowledge	This course will assume knowledge of Analysis 1: NB1201 (or earlier incarnation NB1200, WI1415NB, WI1411NB). Especially the parts on integration techniques and series. You are also expected to have some limited knowledge about vectors (the part which is taught in Dutch high schools: The basic concept, addition and dot products)	
Course Contents	The main part of this course is about functions of more than 1 variable. To do this we start by reviewing and expanding your knowledge on vectors. Subsequently you will learn how to generalize derivatives and integrals to this new setting. You will also learn some basic applications like finding the maximum of a function of more than 1 variable. As a secondary topic we consider Fourier analysis. This is the mathematics behind spectra (for example decomposing music in notes of different frequencies). Finally we will spend some time on describing curves (paths through space) and their properties. Topics of Analysis: vectors, functions of several variables, partial differentiation, multiple integration, Fourier series and transformation, space curves.	
Study Goals	After this course the student is expected to have achieved the following learning goals: 1. The student can calculate with vectors a. evaluate manipulate expressions involving dot and cross products b. describe lines and planes using vectors 2. The student can transform different descriptions of curves (words, equations, parametrisations) into each other and use these to calculate quantitative properties of such curves. a. arc length b. curvature and torsion c. Frenet frame 3. The student can apply the theory of differentiation and integration to functions of two variables: a. The student can visualize functions of two variables (using graphs and level curves). b. The student can calculate partial derivatives c. The student can apply the chain rule for functions of several variables d. The student can find the directional derivative and gradient e. The student can find tangent planes and lines to graphs and level curves/surfaces. f. The student can calculate double integrals (including using polar coordinates) g. The student can determine critical points of a function and their type h. The student can obtain extreme values of a function on several domains (both directly and with Lagrange multipliers) 4. The student can calculate Fourier series and transformations of simple functions, and apply basic properties of these transformations. a. Obtain the coefficients of Fourier (co)(sine) series b. Describe the convergence of Fourier series (including Gibbs effect) c. Obtain a Fourier transformation of a function d. Evaluate a convolution product (and describe the relation with Fourier transformations) e. Rewrite differential equations to an equation for a Fourier transformation. 5. For all the above topics the students can give a few situations (esp. outside mathematics) where it can be applied and is able to apply them to those situations.	
Education Method	(Assuming no new COVID measures) This course will include the following educational activities: - Interactive lectures - Work sessions where you can practice under the guidance of student assistants - Homework to do at home both on computer and on paper Apart from basic questions where you can directly apply the methods learned in class, the homework will also contain questions where you have to be more creative. This will help develop your problem solving skills. (The exam can contain such a more creative question or require one of the techniques you had to discover for yourself in one of the homework problems.)	
Computer Use	Digitally corrected exercises are part of the homework. A digital book is used for the course.	
Literature and Study Materials	- Openstax Calculus volumes 1, 2, and 3. Download the original books for free from openstax.org, or access them online through Brightspace. - There will be some lecture extra notes on Fourier and torsion written by the lecturer The lecture notes are available on Brightspace.	
Assessment	(Assuming no new COVID measures) The (regular) assessment consists of - Two written exams (2x1.5 hours)(Grades M and F)	

	- There are quizzes, which count for 20%. For each week of lectures there is a quiz in the digital exercise system which you can take multiple times. The best 6 out of 8 quiz results count. If you actively participate in at least 6 quizzes, the quiz-grade is taken to be a minimum of 5.75 (so you can't fail due to bad quiz results). Grade Q. The final grade of regular part is thus 2/5 * M + 2/5 * F + 1/5 * Q Students who take this course a second time do not have to do all the quizzes. In that case their grade is 1/2 * M + 1/2 * F. The resit is a single 3 hour exam, for which the quizzes do not count. You have to register in Osiris for both exams and the resit. You do not have to register for the quizzes. Intermediate grades for the regular exam (M, F or Q) cannot be kept for future years. The compensation rule Analysis for Nanobiology is applicable. If the grades for Analysis 1, 2, and 3 (NB1201, NB1206, and NB1211 or prior incarnations) are all at least 5.0 (in decimals), and the weighted (by EC) average of these grades equals at least 5.8 after rounding to decimals, then the student gets the EC for all three courses.
Permitted Materials during Tests	Pen, pencil and a simple, non-graphing calculator. The calculator is not allowed to be able to do calculations with complex numbers or evaluate integrals (either analytically or numerically). As such the Casio fx-570, fx-580, fx-991 and TI-36x are not allowed. Allowed are, for example, Casio fx-82 or TI-30.
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and and may be at a different time of year than the old course.
Studyload/Week	5ECTS = 140 hours total Regular weeks (8 times): 6 hour lecture + 4 hour work session + 4 hour self study = 14 hour/week Exam week (2 times): 12.5 hour exam preparation + 1.5 hour exam = 14 hour/week Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	TU Delft

NB1211	Analysis 3	3
Responsible Instructor	Dr. F.J. van de Bult	
Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4A	
Course Language	English	
Required for	This course will teach some slightly more advanced mathematics which will be relevant to many other courses in the Nanobiology programme. Some examples of courses which rely especially on this course are Physics 1B, 2 (NB1143, NB2141) [Unfortunately due to the scheduling of the courses for Physics 1B this mostly means referring back] Thermodynamics and Transport (NB2011)	
Expected prior knowledge	Analysis 1 and 2: NB1201 + NB1206 (or earlier incarnations: WI1411NB/WI1415NB/NB1200 and WI1412NB/WI1422NB/WI1423NB/NB1205)	
Course Contents	In this course you will learn more about integrals and derivatives of functions of 2 and 3 variables. In particular you will learn about different types integrals over curves and surfaces in space. You will also learn about the corresponding fundamental theorems of Calculus (differentiation and integration cancel each other) related to these integrals. Vector Calculus: change of coordinates for multiple integrals. Line integrals. Surface integrals. Green, Gauss and Stokes.	
Study Goals	After this course the student is expected to have achieved the following learning goals. 1. The student can obtain derivatives of implicitly defined functions (also when defined by multiple equations) 2. The student can evaluate triple integrals. 3. The student can change coordinates in multivariate integrals (more general setting for 2D integrals, to cylindrical and spherical coordinates for 3D integrals). 4. The student can calculate the different types of integrals from 2D and 3D vector calculus using parametrizations a. The student can parametrize implicitly defined curves and surfaces. b. The student can evaluate a line integral of a scalar field on R^2 or R^3 c. The student can evaluate a line integral of a vector field on R^2 or R^3 d. The student can evaluate a surface integral of a scalar field on R^3 e. The student can evaluate a flux integral 5. The student can apply the relevant version of the Fundamental Theorem of Calculus (=generalized Stokes Theorem) to calculate the integrals of study goal 4 a. Fundamental theorem for line integral of a vector field (potentials) b. Green c. Stokes d. Gauss 6. The student can perform calculations using gradient, divergence, and curl a. Evaluate them b. Apply product and chain rules c. Determine (vector) potentials d. Work with second order derivatives (such as Laplacian) 7. For all the above topics the students can give a few situations (esp. outside mathematics) where it can be applied and is able to apply them to those situations.	
Education Method	(Assuming no new COVID measures) This course will include the following educational activities: - Interactive lectures - Work sessions where you can practice under the guidance of student assistants - Homework to do at home both on computer and on paper Apart from basic questions where you can directly apply the methods learned in class, the homework will also contain questions where you have to be more creative. This will help develop your problem solving skills. (The exam can contain such a more creative question or require one of the techniques you had to discover for yourself in one of the homework problems.)	
Computer Use	Digitally corrected exercises are part of the homework. A digital book is used for the course.	
Literature and Study Materials	- Openstax Calculus volumes 1, 2, and 3. Download the original books for free from openstax.org, or access them online through Brightspace. - There will be some lecture extra notes on implicit differentiation written by the lecturer The lecture notes are available on Brightspace.	
Assessment	(Assuming no new COVID measures) The (regular) assessment consists of - one written exam (2 hours) (Grade F) - There is a midterm, which count for 20% (Grade M). - There are quizzes each week. The best six out of 8 quizzes count. The final grade of the regular part is thus 7/10 * F + 1/5 * M + 1/10 * Q Students who take this course again do not have to do the midterm or the quizzes; their grade is then simply F. The resit is a single 2 hour exam, for which the midterm and the quizzes do not count. You have to register in Osiris for midterm, exam and resit. Intermediate grades for the regular exam (F, M or Q) cannot be kept for future years. The compensation rule Analysis for Nanobiology is applicable. If the grades for Analysis 1, 2, and 3 (NB1201, NB1206, and NB1211 or prior incarnations) are all at least 5.0 (in decimals), and the weighted (by EC) average of these grades equals at least	

	5.8 after rounding to decimals, then the student gets the EC for all three courses.
Permitted Materials during Tests	Pen, pencil, and a simple, non-graphing calculator. The calculator is not allowed to be able to do calculations with complex numbers or evaluate integrals (either analytically or numerically). As such the Casio fx-570, fx-580, fx-991 and TI-36x are not allowed. Allowed are, for example, Casio fx-82 or TI-30.
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and and may be at a different time of year than the old course.
Studyload/Week	3ECTS = 84 hours Regular weeks (8 times) 4 hour lecture + 2 hour work session + 2 hour self study = 8 hour/week Midterm (once) 7 hour preparation + 1 hour midterm = 8 hour/week Final (once) 10 hour preparation + 2 hour final = 12 hour/week Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	TU Delft

NB1230	Linear Algebra	3
Responsible Instructor	Dr. I.M. Smit	
Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Required for	Linear Algebra is an important mathematical skill and language that will be used throughout the program. In particular, Linear Algebra plays a significant role in the courses on Differential Equations (NB2191), Physics 2 (NB2141),Systems and Signals (TN2545) Image Analysis (NB2121), and Quantum Mechanics (NB3017,NB3018), .	
Expected prior knowledge	- High school mathematics B, or equivalent. - Analysis 2 (NB1205) (the part on vectors).	
Course Contents	Linear Algebra is a branch of mathematics which is based on two mathematical operations: addition and (scalar) multiplication. Despite this deceptive simplicity, it has a vast variety of applications, both within and outside mathematics. This course is an introduction to several fundamental aspects of Linear Algebra, like vector calculation, solving systems of linear equations and matrix algebra, and notions like linear combinations, linear (in)dependence, linear transformations, subspaces, bases, dimension, eigenvalues and eigenvectors, diagonalisation, standard inner products, orthogonal projections and least squares methods.	
Study Goals	The most important learning objectives for this course are given below. The student 1. can solve systems of linear equations and make a priori statements about the number of solutions; 2. can perform matrix operations (addition, multiplication, inverse, determinant) and can determine the matrix of a linear transformation; 3. can determine whether a set of vectors is linearly dependent or linearly independent, and relate this to other concepts in linear algebra; 4. can determine dimension and bases of linear subspaces, in particular of the null space and column space of a matrix; 5. can find the (possibly complex) eigenspaces and eigenvalues of a given matrix; 6. can determine whether a matrix is diagonalisable or not, can diagonalise a matrix if possible and use this to analyse dynamical systems; 7. can find orthogonal decomposition of a vector with respect to a linear subspace; 8. can find least square solutions of a system of linear equations, and use this to fit curves to data. Please note: the student should know and understand all terms mentioned, as well as the notions that are implicitly needed to comply with the learning objectives.	
Education Method	- In a general week, there are four hours of interactive lectures ("colstructies"), in which the teacher presents new material. There will also be time to work on and discuss exercises. - In addition, there are 2 hours of tutorial classes, in which there is time to work on homework under supervision of the teacher and teaching assistants. These hours will also be used to discuss some of the more challenging exercises. - There are two bonus quizzes. - Homework consists of computation-oriented exercises in Grasple and more conceptual exercises provided on Brightspace in pdf format. Your education in linear algebra will help you build your professional skills in problem solving, writing (in the language of linear algebra) and team work. Even though your exam is individual, cooperating during exercise class will help you learn the material more thoroughly. * Due to numerous holidays in Q4, schedules might be slightly different week-to-week.	
Literature and Study Materials	- Book: Lay, Linear Algebra and Its Applications, Global Edition, 6th edition, ISBN 978-1292351216. Using an older edition of this book is no problem. This book can be obtained through the student association or StudyStore (www.studystore.nl). - Grasple: An online homework environment that you can access through Brightspace. - Problem sets on PDF, available on Brightspace.	
Assessment	The assessment consists of: - Two bonus quizzes (~30 minutes each) for in-between feedback on your progress. - A final 2-hour exam. Part of this exam may be a Bring-Your-Own-Device Grasple test, and part of this exam will be on paper. The final grade equals the exam grade, but the quizzes can give you a bonus: if the average quiz grade is higher than the exam grade, then the exam grade is increased by 20% of the grade difference, with a maximum of 0.5 point. In formula: F=E+min{0.5,max{0,0.2(Q-E)}}, where F is the final grade, E the exam grade and Q the average quiz grade.You passed the course whenever the final grade is at least 5.75. The resit consists of a 2-hour exam (possibly a Bring-Your-Own-Device / paper combination). Note that if you take the resit exam, the aforementioned quizzes will not contribute to your grade. Registration for exams is done through Osiris.	
Permitted Materials during Tests	Pen and paper. If applicable: Laptop for Bring-Your-Own-Device exam part.	

Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments (such as quizzes), students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The Nanobiology Bachelor programme is undergoing a major curriculum restructuring. The new curriculum will begin for first year courses in 2025-2026. 2024-2025 is the last time this course for this number of credits will be offered. There will be 2 resit opportunities in 2025-2026 and transition rules for students who don't complete them. The resits in 2025-2026 will likely be at different times of year. The new courses will often combine courses and and may be at a different time of year than the old course.
Studyload/Week	The total study load is 3 ECTS = 84 hours. I would suggest to distribute these as follows: - 16 x 2 equals 32 hours of lectures. - 8 x 2 = 16 hours of exercise class. - 8 x 2.5 = 20 hours on homework - 1 hour quiz time - 13 hours exam preparation - 2 hours exam time Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	Delft

Year	**2024/2025**
Organization	**Applied Sciences**
Education	**Bachelor Nanobiology**

2e jaar BSc NB 2024	
Responsible Program Employee	J.C. Colgrove

NB2011	Thermodynamics and Transport	3
Responsible Instructor	Dr. P. Boukany	
Responsible Instructor	Dr. B. Bera	
Instructor	Dr. B. Bera	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course deals with transport of mass (or moles), and energy - the basic quantities in classical physics - at the level of cell. Most processes involve transport- or transfer in general - of these quantities. Mass, fluid and heat transport processes are fundamental to virtually all aspect of life. For example, living systems need energy to survive. Moreover, they need new material, in the form of specific molecules to grow and reproduce or get their energy from. Thermodynamics: -First law of thermodynamics; Equation of States; Changing States -Entropy and second law of thermodynamics; Reversible process and work/heat; -Entropy, Enthalpy and Gibbs Free Energy; Chemical potential Transport phenomena: -Microscopic theory of Diffusion; Brownian motion; Random Walks -Macroscopic description of diffusion; Fick's equation; Diffusion through pores -Mass transport across a thin film; Reaction-Diffusion system; -Life at low Reynolds number; Diffusion with drift; Stokes flow; Flow through a channel; Sedimentation flow Mathematical analysis methods: scaling and approximation techniques, analytical and numerical approaches.	
Study Goals	After this course the student should: 1. have knowledge of the basic definitions and concepts of thermodynamics and transport phenomena at the scale of the cell; 2. be able to analyse basic problems concerning heat, mass and momentum transfer in and around (living) cells; 3. be able to model the related processes and solve the models to obtain a quantitative description of the situation; be able to assess the outcome of the analysis using a first principle approach.	
Education Method	Class room lectures followed by exercises and guided self study. Note: The exact format of course may be changed due to Coronavirus (COVID-19) regulations.	
Books	Random walks in biology, Howard C. Berg	
Prerequisites	BSc: Physics 1 Note: it is absolutely required to have a good working understanding of BSc level Mathematics.	
Assessment	Mid term exam (Thermo) and final exam (Transport): both are of the problem solving type. In addition, during the course two assignments will be handed out. First assignment is related to thermodynamics, and includes the results from a practicum. The second assignment is related to transport phenomena. These assignments are graded; the grading forms part of the final mark. Final mark = (10%) assignment-1 (Thermo) + (10%) assignment-2 (Trans) + (40%) midterm exam (Thermo) + (40%) final exam (Trans) The grades for the assignment are also valid for the next academic year. That means you can keep the assignment grades if you are following the course for the second year in a row. In case you have to resit, only the exams can be resit, and only a combined resit exam covering both topics can be taken. Note: The exact format of assessment may be changed due to Coronavirus (COVID-19) regulations.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Studyload/Week	6h/week*8week=48h (lectures+working class) 3h/week study (3*8 week=24 hr) time at home 12 h exam preparation	
Location	Delft	

NB2022	Philosophy and Ethics	3
Responsible Instructor	M.M. Jaegle	
Instructor	Dr. J. Pothof	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 2B 3A	
Course Language	English	
Course Contents	In this course the student will be introduced to some key aspects of scientific integrity and ethical issues. The understanding gained in this course is needed to appreciate how the knowledge learned in other courses such as molecular biology and biophysics was generated and to appreciate that this knowledge may change. Explicit discussion of scientific and professional ethics develops a students understanding of their contribution to a cooperative and productive scientific community. Literature and study material: * Reading materials provided by the lecturers	
Study Goals	At the completion of this course the student will be able to: 1. demonstrate understanding of ethical behavior in research; 2. demonstrate knowledge and understanding of important ethical values and principles associated within scientific research; 3. reflect on ethical issues that arise in scientific research; 4. discuss ethical questions with peers and scientists.	
Education Method	Education methods: * Lectures and discussion sessions. * Group assignments and individual assignment Professional skills: - Group Presentation - Team work - Writing report - Strong Work Ethic	
Literature and Study Materials	* Reading materials provided by the lecturers * Online lectures provided by the lecturer	
Books		
Assessment	There are 3 required elements. All must be passed separately to receive a pass in the course. 1-Mandatory attendance at all the sessions. (Pass/fail) 2-Weekly individual assignments involving: The writing of a short reflection essay on the lecture topic of the week. (Pass/fail) A case study. (Pass/fail) You are permitted to miss one session and associated assignments. 3-Group assignment "Interview with a scientist" includes written report: (Pass/fail) Re-sit: - Students who fail the group assignment, may submit a revised version prior to the re-sit week. - Students who fail to attend two sessions or do not write and submit two individual assignments on time will have to re-sit. Missing 3 or more sessions and associated assignments (including late submissions) is considered a fail, and students will need to redo the course next year.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	This course will be taught all of Q2 beginning in 2023-2024	
Remarks	Martine Jaegle can be reached at m.jaegle@erasmusmc.nl	
Studyload/Week	3/week (lectures and discussions) 5/week homework and weekly assignments 16h for interview assignment Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Erasmus MC, teacher can be reached at m.jaegle@erasmusmc.nl	

NB2032	Evolutionary Developmental Biology	6
Responsible Instructor	Dr. H. Mira Bontenbal	
Instructor	Dr. B.C. Tilly	
Instructor	Dr.ir. W.M. Baarends	
Contact Hours / Week x/x/x/x	0/0/7/0	
Education Period	3 3A 3B	
Start Education	3A	
Exam Period	3 3B 4A	
Course Language	English	
Course Contents	All animal species develop from a single cell, the fertilized egg, and share genes which regulate development. A large number of genes encoding components of molecular and cellular mechanisms involved in development are highly conserved among animal species. Yet, nature shows us an astonishing variety of animal forms. The basis for this variety lies in genome diversification through mechanisms such as gene duplication and mutation events, which generate opportunities for further modification and refinement of developmental schemes, resulting in new forms and functions, if the new phenotypes are successful under natural selection. Evolution and diversification of animal forms occurs through changes in development. By studying developmental biology in an evolutionary context, we gain a much better understanding of the mechanisms at work in animal development. In this course, the students will learn about general aspects of development across the animal kingdom: from worms and flies to fish, birds, and mammals. It goes without saying that this approach also provides an illuminating picture of our own development.	
Study Goals	Upon completion of this course, the student will be able to: 1. Describe evolutionary conserved concepts of animal development and cell differentiation in various model species (Understand) 2. Identify embryo stages and key structures at different phases in different model organisms (Remember) 3. Explain similarities and differences between different model species at the level of gene regulation, cells, embryo structures, organs and limbs; using knowledge on the biological processes that are at the base of those similarities and differences. (Understand) 4. Describe the basic evolutionary concepts about the development of multicellularity, sexual reproduction, and Darwins theories concerning speciation (Understand) 5. Connect specific evolutionary events to the appropriate evolutionary concept. (Apply) 6. Provide the definitions of toti- and pluripotency of cells (Remember) 7. Assess the possibilities of the use of different types of (induced) pluri- and totipotent stem cells in research and therapeutic applications (Evaluate) 8. Critically interpret the results of scientific experiments in developmental biology, using provided text and figures from scientific manuscripts. (Analyse) 9. Develop a research question using information gained from a scientific literature search (Create) 10. Assess the literature on the poster topic and summarize the findings in components of a poster (Evaluate) 11. Constructively collaborate with fellow students to prepare a scientific poster, and presentation (not applicable, it is an attitude type of learning aim) 12. Present and defend a scientific poster to peers in an engaging and comprehensive manner (not applicable, it is an attitude type of learning aim) 13. Critically evaluate scientific presentations of peers using an assessment rubric (Evaluate)	
Education Method	- interactive lectures - interactive practicals - integrative sessions where non-graded homework are corrected and non-graded quizzes are done by the students - homework (non graded) - workshops, meeting with C. elegans and zebra fish - visit to the Natural History Museum Rotterdam - assignment: make a poster (shared responsibility of 2-4 students); these posters will be presented by the students at a poster session, whereby the teachers and the students will assess the quality of each poster using an evaluation form. This will develop your soft skills: group work, poster presentation and peer review.	
Literature and Study Materials	- Principles of Development; Wolpert and Tickle; Sixth Edition 2019 - Students should already have: Molecular Biology of the Cell; Alberts et al.; Sixth Edition 2015 - Open Access chapters, will be provided by teacher.	
Assessment	The final grade consists of three parts - Part 1: Midterm exam (open questions and multiple choice questions), passing grade is 5.8 or higher. Electronic exam. This will cover the 1st part of the course. - Part 2: Final exam (open questions and multiple choice questions), passing grade is 5.8 or higher. Electronic exam. This will only cover the 2nd part of the course, not the full course. - Part 3: Poster Presentation, passing grade is 5.8 or higher. The global course grade is an average of each parts grade (1/3 Part 1, 1/3 Part 2, 1/3 Part 3). Parts 1 and 2 can compensate for each other but Part 3 cannot compensate. This means the student will have to pass the exam (average of Part 1 and Part 2) and Part 3. The grade for Part 3 will be kept until the exam has been passed. The ECTS for this course will only be assigned to you when the online course "Information Literacy 2" is successfully completed. The exams will take place electronically, while the poster session will be physical. In terms of resits: - At the exam resit, the student can chose to do Part 1 or Part 2, or both. The highest grade of each Part will be kept for the global course grade. - There is no resit for Part 3. - The grades of Part 1 or Part 2 cannot be kept for a subsequent year: the student will have to redo both parts if the average grade is not equal or above 5.8.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration.	

	Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	This course will be taught in Q3 beginning in 2022-2023.
Remarks	Hegias Mira Bontenbal can be reached at h.mirabontenbal@erasmusmc.nl Willy Baarends at w.baarends@erasmusmc.nl
Studyload/Week	6ECTS, 2 octals, 6ECTS = 168h 7h/week * 8 weeks = 56h (lectures and working classes) Leaves ca. 10h/week study time at home, homework, poster preparation + 32h exam preparation Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Schedule	We will put our up-to-date schedule on Brightspace. So please follow that schedule and not the schedule on mytimetable.tudelft.nl. Use the Brightspace one when the course starts.
Location	Rotterdam

NB2071	Physical Biology of the Cell 2	3
Responsible Instructor	D.H.M. Meijer	
Responsible Instructor	Prof.dr. G.H. Koenderink	
Instructor	Prof.dr. G.H. Koenderink	
Contact Hours / Week x/x/x/x	0/5/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for understanding the cellular basis of development (course Evolutionary & Developmental Biology).	
Expected prior knowledge	The course will assume knowledge of cellular biomolecules gained in the courses of Chemistry, Molecular Biology and Genetics. It partly builds on concepts introduced in Physical Biology of the Cell 1.	
Course Contents	Biological cells are complex biochemical systems that obey the laws of physics. Modern research in molecular and cell biology therefore increasingly relies on physical concepts. In Physical Biology of the Cell 1 you learned how to utilise such concepts to model cellular phenotypes on a macroscopic scale. In Physical Biology of the Cell 2 you will transfer this approach to the molecular scale and learn how signalling cascades, and (experimental) models to interpret them, are essential as we build our molecular understanding of cellular phenomena such as information processing, translocation across membranes, and mechanobiological signalling. In this course you will learn how to connect concepts from biology and biophysics with observations from biochemistry and physics to describe this diversity of cellular processes and to understand their molecular basis. You will then apply these concepts to concrete examples in which you will establish quantitative models for the regulation of biochemical cascades and circuits, the functional dynamics of molecular machines and processes related to translocation and the propagation and processing of biological signals.	
Study Goals	Upon completion of this course you will be able to: 1. Identify fundamental challenges in cellular information processing 2. Describe how protein-protein interactions are used to transmit information 3. Understand how canonical classes of signaling proteins couple conformational change with catalytic activity 4. Understand how changes in post-translational modification and/or subcellular localization of signaling molecules transmit information 5. Distinguish functional differences in speed, efficiency and specificity of second messengers 6. Understand structural elements and organization of the cell membrane 7. Explain the function of mechanobiological signaling in different tissues 8. Apply concepts of cell signaling to model signaling pathways 9. Interpret schematic information from a figure to both a biophysical and a biological meaning. 10. Relate current open questions in cellular information processing to experimental approaches 11. Evaluate primary literature on novel technologies to study cellular communication	
Education Method	Lectures (plenary) Presentations (teams) Assignments (in subgroups)	
Literature and Study Materials	Required textbooks: 1. Cell Signaling. Principles and mechanisms, Lim, Mayer, Pawson 2. Molecular Biology of THE CELL, Alberts	
Assessment	Students will be assessed according to the study goals as formulated above. The exam consists of open questions probing your quantitative and qualitative understanding of the study goals. The exam will be conducted in a computerized format. To pass the course you need at least a 5.5 for the exam. Please note that you will need to score at least a 5.8 for your final grade (assignments, presentation and exam combined) to pass the course. Grading Track 1 Final exam (70% of the final grade, individual) Presentation (20% of the final grade, teams) Homework assignments (10% of the final grade, individual) Homework assignments will be assessed on a pass/fail scale. Track 2 Final exam (80% of the final grade, individual) Presentation (20% of the final grade, teams) Students will decide between track 1 and track 2 within two weeks after the start of the course.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	This course will be taught in Q2 beginning in 2022-2023	
Studyload/Week	3 ECTS, octal course = 84 h 5h/week * 8 weeks = 40h (lectures and working classes) Leaves ca. 10h/week study time at home + exam preparation Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	

NB2081	Nanotechnology	2
Responsible Instructor	Dr. M.E. Aubin-Tam	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Prior to the beginning of this course students are expected to have completed the first year lab courses, and the courses Physics 1A (NB1140), Physics 1B (NB1143), Biophysics (NB1132), and Physics 2 (NB2141).	
Course Contents	In this course the student are introduced to current and developing technology used in Nanobiology research. They will learn the basic principles of instruments such as optical and magnetic tweezers and atomic force microscopy. The use of nanofabrication, microfabrication and microfluidics in nanobiology will be illustrated. The type of data produced, and the relevant sensitivity and accuracy of the different instruments will be described. Students will work hands-on with one of these instruments. The relevant application of specific instruments to answering questions in biology will be discussed. Bottom-up and top-down nanofabrication approaches will be described. This course will assume knowledge of Physics, Math, Biophysics, and Molecular Biology as learned in YR1 and previous YR2 courses. Knowledge learned in this course is needed to understand current advances in nanobiology and make informed choices for thesis project work.	
Study Goals	At the completion of this course the student will be able to: 1. Accurately describe the working principle of several (at least 3) instruments currently used in nanobiology research 2. Identify the type of data each instrument can produce and name relevant applications in nanobiology research 3. Describe how one such instrument is calibrated 4. Identify areas of application in medicine that have benefited form nanotechnology instrumentation 5. Describe bottom-up and top-down nanofabrication approaches.	
Education Method	Interactive lectures Self / group study practicum and report writing. This will develop these skills: taking safety measures when doing practical work, scientific writing, team work. Practical work with selected instrument	
Assessment	Assignments: 60% ; Final exam: 40% ; Note that the retake exam only allows to change the grade of the final exam, while the grade of the assignments cannot be changed. Participation to all the practical sessions is required to pass the course.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments. Students must enrol in the course in Brightspace no later than the 22nd of April 2025, so that they can be scheduled for their practicum.	
Studyload/Week	2EC = 56h; 6 lectures of 2h = 12h; 2 practicum = 32h (2h of preparation, 8h of practical work, 22h for data analysis and writing); 12h for exam preparation. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation, and in no way this guarantees how much work it will take for you.	
Location	Delft	

NB2141	Physics 2	3
Responsible Instructor	Prof.dr. C. Joo	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 2A	
Course Language	English	
Course Contents	In this course, students will receive an introductory overview of modern physics, including optics, special relativity, and quantum mechanics, among other topics. These physical principles will be taught through the lens of biological examples, providing a unique perspective on the relationship between physics and the natural world.	
Study Goals	Upon completion of this course, students will have gained the following skills and knowledge: 1. A comprehensive understanding of the fundamental laws governing optics, special relativity, and quantum mechanics. 2. The ability to describe physical phenomena using mathematical concepts. 3. Insight into how optics and quantum mechanics are used in the field of biophysics research. 4. The ability to apply the basic physical concepts learned in this course to more advanced topics in biology, biophysics, and physics. This includes subjects such as optics, image processing and data analysis, statistical physics, biophysics, physical biology of the cell, and introduction to nanotechnology. Overall, students completing this course will have a solid foundation in modern physics principles and how they can be applied to biological systems, opening up exciting possibilities for further study and research in related fields.	
Education Method	This course incorporates integrated lecture and exercise sessions that provide a comprehensive learning experience. The lectures cover all chapters of Wolfson's textbook on optics, special relativity, and quantum mechanics. At the start of the course, a detailed overview of the material to be covered in each lecture will be provided, allowing students to prepare and engage with the material effectively. The exercise sessions provide an opportunity for students to apply the concepts learned in the lectures through problem-solving and group discussion, further reinforcing their understanding of the material.	
Literature and Study Materials	Required: Essential University Physics, Richard Wolfson, Addison-Wesley, and Mastering Physics, Richard Wolfson, Addison-Wesley. Supplementary: Principles of Physics, David Halliday, Robert Resnick and Jearl Walker, Wiley, 2011.	
Prerequisites	Knowledge of calculus (Analysis 1, 2 and 3), Physics 1, Biophysics	
Assessment	Mid-term exam (50%, 90 minutes) Final exam (50%, 90 minutes) Resit exam (180 minutes) is a combination of the two exams from the course	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Studyload/Week	This course is worth 3 ECTS, which corresponds to approximately 84 hours of work. Over the course of eight weeks, students can expect to spend around 6 hours per week attending lectures and exercise classes, totaling approximately 44 hours. This leaves approximately 4 hours per week for independent study and review at home, as well as an additional 10 hours for exam preparation. It is important to note that every student is different and may require more or less time to master different topics. This is a rough estimate and should not be taken as a guarantee of how much work each individual student will need to put in to succeed in the course.	
Location	Delft	

NB2161	Bioinformatics	4.5
Responsible Instructor	Dr. J.A. Baaijens	
Instructor	A. Mahfouz	
Instructor	Dr.ir. S.J.J. Brouns	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Bioinformatics is increasingly important to help answer biological questions, to deal with large volumes of measurement data which are now easily generated in the lab in the light of the vast amounts of prior knowledge already present in biological literature and stored in databases. In this course, the student will learn how to use computer methods as tools for biological research. An overview of core application software, databases and online applications will be provided, as well as in-depth knowledge on key algorithms and procedures to process and analyze sequence data and quantitative measurements. The course offers both theoretical insight and practical experience, in computer practicals and projects to be performed.	
Study Goals	At the completion of this course, you should be able to: 1. Describe a range of bioinformatics tasks, tools and databases 2. Explain in detail the core bioinformatics algorithms for sequence alignment, statistical testing, clustering and classification 3. Argue which algorithms/tools/databases to use for solving a given biological data analysis problem 4. Apply bioinformatics algorithms/tools to real biological data 5. Interpret the results from a bioinformatics workflow	
Education Method	- Lectures - Computer practicals - Projects	
Books	We will use two books, both of which are publicly available online: - Modern Statistics for Modern Biology, by Susan Holmes and Wolfgang Huber (https://www.huber.embl.de/msmb/) - Bioinformatics Algorithms, by Philip Compeau and Pavel Pevzner (https://www.bioinformaticsalgorithms.org/read-the-book)	
Assessment	The course grade is calculated by a weighted average of the Exam grade (60%) and the Project grade (40%). - Exam (60% final grade): One exam (3 hours) covering all the modules of the course. - Projects (40% of final grade): The projects are graded on a scale 0-10 and require you to hand in a deliverable and present your results. In addition, there are practicals: Practicals are available per module to practice the material and need to be completed before the given deadline. The practicals are graded pass/fail, where an overall pass (60% of total points) is needed to participate in the final exam and resit. The deadlines for the practicals and project deliverables will be presented in the first lecture and are presented in the course schedule in the electronic learning environment. Late policy: Late submission of a project results in a daily -1 (minus one) on the project grade for every day from the submission deadline until submission. Notes: - No scores and/or submissions are transferred to the next year. - In order to pass the course, students must score at least 5.0 on the exam (minimal passing grade). - The resit grade only replaces the exam grade (60% of final grade), not the project grades. - There are no repair opportunities for late submissions (projects and practicals) - Exceptions to the late submission policy are possible based on recommendation by the academic counsellor.	
Permitted Materials during Tests	Calculator	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Studyload/Week	NB2161: 4.5 ECTS, 2 octals 4.5 ECTS = 128h The course has 56 contact hours (lectures and working classes). This leaves ca. 5h/week study time at home + 32h exam preparation. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Delft	

NB2171	Statistics	3
Responsible Instructor	Dr. E. Pulvirenti	
Instructor	Prof.dr. F.H.J. Redig	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course covers the basics of probability theory and statistics. Topics in probability include discrete and continuous random variables, probability distributions, and stochastic independence, joint distributions, sums and functions of random variables, expectation, variance and covariance, law of large numbers, and central limit theorem. Topics in statistics include random sample and sample distributions, estimators (bias, error, and maximum likelihood), least squares and confidence intervals for the average of normally distributed data with known or unknown variance, and hypothesis tests.	
Study Goals	By the end of the course, you should be able to: 1. compute conditional probabilities and apply the notion of independence 2. use discrete and continuous random variables and calculate expectation and variance 3. apply the law of large numbers and the central limit theorem 4. analyse data by means of graphical and numerical summaries 5. compute the efficiency of an estimator, evaluate whether it is biased and compare different estimators 6. compute the maximum likelihood estimator 7. construct confidence intervals 8. perform hypothesis tests and compute errors of the first and second kind	
Education Method	Lectures and exercise classes	
Books	A Modern Introduction to Probability and Statistics Understanding Why and How Springer Texts in Statistics, Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover, ISBN: 1-85233-896-2	
Assessment	On campus digital exam, 2 hours exam. It may be a BYOD (bring your own device) or in a computer testing hall depending on scheduling.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Studyload/Week	3ECTS = 84h 16 lectures * 2h/lecture = 32h (lectures) study pre-lecture and pre-lecture exercises 1h -> (15h in total) Leaves about 20h study time and exercises from lectures and 20h exam preparation Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	TU Delft	

NB2181	Computational Science	3
Responsible Instructor	Ir. M.J.F. Negrello	
Instructor	N. Šoštari	
Contact Hours / Week x/x/x/x	0/0/0/0/4/4/4/0	
Education Period	3 4 3A 3B 4A	
Start Education	3	
Exam Period	4	
Course Language	English	
Required for	This course delivers essential knowledge for all scientific computing courses in the nanobiology program and in a professional career.	
Course Contents	The student learns to use modelling and simulation techniques in science and engineering, with special emphasis on nanobiology. The students first become familiar and/or improve their existing skills with the programming environment of Python. The students work then on assignments (individually or in groups) where they learn different numerical methods and their implementation in Python. At the end of the course the students use computer modelling in order to analyse and solve a scientific problem related to nanobiology.	
Study Goals	At the end of this course 1. The student can analyze, manipulate and visualize data in the python programming language, and specifically, in the jupyter notebook environment. 2. The student can translate mathematical statements and physical models into numerical algorithms into working computer code. 3. The student can reason about the effect of data resolution, numerical precision, convergence, computer memory and computational speed on the numerical solution of a scientific or engineering problem. 4. The student can combine computational tools in linear algebra, random numbers, calculus into a computational model of physical phenomena.	
Education Method	- Practical activity where students solve python projects on basic skills necessary for scientific computing. Skills acquired: - Translating mathematical models into computational algorithms - Debugging - Explaining own code - Commenting code	
Assessment	Normal assessment: An electronic exam worth 60% of the mark and a final project with oral assessment worth 40% (both have separate resits). In the case of off-campus exams, the assessments may be modified.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The teacher can be reached at m.negrello@erasmusmc.nl	
Studyload/Week	In addition to lecture hours, students are expected to invest an extra 4h/week honing their coding skills. Students with little to no previous coding knowledge should expect to invest up to 8h/week. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	TU Delft	

NB2191	Differential equations	3
Responsible Instructor	Dr.ir. Y.M. Dijkstra	
Responsible Instructor	Dr. W.M. Schouten-Straatman	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2 1A 1B 2A	
Course Language	English	
Course Contents	Differential equations occur in all branches of science and engineering as mathematical models for reality. Answering questions that may pop up in any of these branches often comes down to analysing and solving such differential equations. The aim of this course is to introduce you to a variety of techniques to solve differential equations of several commons types. In addition, we will discuss techniques to extract information from differential equations without solving them.	
Study Goals	1. For a given (system of) differential equations the student can tell whether it is linear / non-linear, homogeneous / non-homogeneous, separable, exact or partial. 2. The student can make statements about existence, uniqueness and analyticity of solutions for boundary value problems and initial value problems without explicitly solving. 3. the student can make statements about qualitative properties of solutions such as limiting behavior, behavior near equilibrium points and radius of convergence of power series, without explicitly solving. 4. The student can apply various methods for constructing solutions to (a system of) DEs: Separation, variation of constants, reduction of order, using potentials, Laplace transform, series solutions, Fourier series.	
Education Method	Lectures (4 hours per week) and instruction classes (2 hours per week). In principle all activities will be on campus.	
Books	William E. Boyce & Richard C DiPrima, the twelfth edtition, international adaptation ISBN: 9781119820512	
Assessment	One midterm (40%) and endterm(60%), no bonus regulation. Details will be provided at the beginning of the course. There are no separate resits for midterm and endterm, only one resit for the full course with weight 100%.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Studyload/Week	This is a 3EC course, which comes down to 84 hours. Contact hours: 48 hours (16 2-hour lectures + 8 2-hour exercise classes) Homework (outside exercise classes): 20 hours (2.5 hours per week, 8 weeks) Exams: 16 hours (5 hours exams, 11 hours preparation) Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Delft	

NB2214	Electronic Instrumentation	6
Responsible Instructor	Dr.ir. M.W. Docter	
Instructor	Ing. J. Bastemeijer	
Contact Hours / Week x/x/x/x	6/6/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	the Python programming will be continued in Bioinformatics NB2161 and Computational Science NB2181. the Fourier transforms and Aliasing will be more quantitatively explained in Signals and Systems TN2545. The ability to use instrumentation and know how it influences your data will be relevant for all future data acquisition and analysis.	
Expected prior knowledge	- Pre-knowledge from the first year Nanobiology are Physics 1b NB1143 (electromagnetism), and BioPhysics NB1132 (Python programming) - It is recommended to follow Signals and Systems TN2545 and Differential Equations NB2191 at the same time as the Electronics Instrumentation course. Some concepts as Fourier Domain and Aliasing are treated qualitatively in EI, and quantitatively in S&S. You will need differential equations to derive some specific circuit behaviour.	
Course Contents	Each week consists of the following components: - Electronics theory - Predictions about how students expect it to work - Programming and/or simulations - Building Circuits or devices and taking measurements - Evaluating results of all three prior components. Octal 1: Passive Electronics: - Learn the electronic theory behind passive components (resistor(s), capacitor(s), inductor(s)), individually and in a circuit (i.e. Thevenin equivalents, filtering) - Learn how to use electronic equipment as an oscilloscope, function generator and ALPACA. - Predict how the combination of such components results in a -frequency (in)dependent- output voltage or current - Verify your prediction with an LTSpice simulation - Build these circuits with the classroom equipment and ALPACA: explore their behaviour by slightly altering the circuit, and analyse the results in time and Fourier domain - Analyse and evaluate your predictions, simulations, measurement and analysis and come to a consistent conclusion, which also takes into account the effect of instrumentation on your data Octal 2: Programming & Data Acquisition: - Develop programming skills in Python (calculations, for loop, if statement, comparison, functions, plotting) - Write programs to control instruments - Acquire, record and analyze (Analog & Digital I/O) data via a computer - Understand, control and measure Aliasing, using anti-aliasing and/or digital filtering Octal 3: Active Electronics: - Understand the electronic theory behind directional components (e.g. a diode) and active components (operational amplifiers, a.k.a. opamps). - Design, simulate and build diode circuits, amplifiers, sensors, bridge circuits and instrumentation amplifiers - Understand and adjust for non-ideal behavior of diodes and opamps - Understand and adjust for noise & interference Octal 4: Active Electronics continued & Project: - Continue the use of sensors - Design, simulate and build an oscillator. - Design and construct a measuring device (for voltammetry)	
Study Goals	After successful completion of this course students are able to: 1. Predict the behaviour of a circuit. 1a. Understand the working mechanism of basic components (passive and active, and their use in filters, sensors, amplifiers , etc) 1b. Understand the most important causes of noise and interference, and to analyse the influence of these on a measured signal 1c. Understand the influences of the electronic equipment on their measurement outcome 2. Design simulations of electronic circuits 3. Design, basic electronic circuits, and measure their activity by hand or via computer 3a. Use basic electronic measuring equipment and data acquisition software: control instruments using a computer interface. 3b. Be able to use a computer to collect data and analyse these data 3c. Know how to optimize the quality of the measured signal. 4. Evaluate the predictions and experimental results, solve possible differences between them 4a. Be able to communicate and collaborate with electronics experts to design good measuring systems.	
Education Method	The education method consists of interactive tutorials, practicums and homework. There will be individual work and group work. In octal 4 students do two projects, 1. (one week) Redesigning the oscillator, 2. (3 week) Designing a measurement device. 2 hours interactive tutorials (theory+predictions) per week, 4 hours Practicum sessions using electronics set-up per week For every theory session there is material to study before the session. After the theory, you will have required preparations (predictions&simulations) to make before each practicum.	
Computer Use	Students will use two computers, the computer in the studio classroom and their own laptop. They must bring their laptop to the theory and practicum every time. - For use of the ALPACA (electronics board), students connect the ALPACA to their laptop with an USB 2 connector, students should bring an adapter if necessary.	

	- Students will need to install a local copy of Anaconda, Jupyter Notebook, and the Alpaca kernel on their laptop. Some assignments include taking photos of circuit boards and layouts, students should bring their smart phone with them so they can do this.
Literature and Study Materials	The practicum manual is available as -online- Jupyter Notebooks (via BrightSpace, Vocareum). The slides, videos and other additional materials are provided through BrightSpace. Book: Electronics: A Systems Approach (6th edition), author: Neil Storey, Pearson Education Limited
Assessment	Participation in practicums is obligatory. - You can only join an exam after finishing all corresponding practicums. - There will be one practicum catch-up session each octal scheduled during the exam week (1.5, 1.10, 2.5) for students who have missed a practicum. - All practicum work requires individual preparation. - The build, measure and evaluate part of the practicum work is done in teams of 2 students, as is the final project and report. The teams for the practicum are formed at the 1st practicum session; the teams for the final project are formed in week 2.6, these teams may be the same. Students may request to do it as a solo activity or be required to do it solo in extenuating circumstances. - Students must indicate via the method described in Brightspace who they will do their final report with no later than the end of week 2.8. This will constitute registration for this interim exam. Again, students may request to do it as a solo activity or be required to do it solo in extenuating circumstances. Students must request this at the time of registration. - The final project report cannot be resubmitted, and has a strict deadline. The final grade is an average of 4 interim examinations: 3 written exams and a report, each counts for 25% of the final grade. You pass the course when your overall score is 5.8 or higher. - Individual exam grades need to be a 5 or above - Completion of practicums may carry forward to the next two academic years, but only when all practicum exercises of the full course, including final project, are completely finished. A retaking student who did not finish all practicum exercises the year before, has to (re)do all practicums. Exceptions are handled by the study advisor. Exam structure: - Three written exams done on computers: 1.5 hour exams in week 1.5, 1.10, 2.5. - Each interim exam has a separate resit exam. Usually 5 weeks after the initial exam. The resit schedule is available in MyTimeTable - The report is on the final project, the deadline is the end of week 2.10, this is done in teams of 2. transition regulation (if you have recorded grades and/or completed the practicum in 2021-2023): rules have changed in 2023-2024. The most beneficial rules apply, if you started the course in 2021 or 2022.
Permitted Materials during Tests	The written exams are digital, meaning they are on a TUDelft computer on campus. Students may only use the files available to them in the testing environment. You are provided with: - computer (including questions, formula sheet and calculator) - scrap paper You can bring with you: - calculator (simple one) - pencil It is a closed book exam. The final report is the product of two people working together as a team, and must include a description of who did what and resources used.
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Studyload/Week	6ECTS = 168h 16 weeks * 2 hours lectures and instruction= 32 h 16 weeks * 4 hours practicum = 64 h 16 weeks * 2 hours homework and study to prepare for theory classes and do predictions =32 h 3 exams * 1.5 hours exam = 4.5 h 3 exams * 6 hours preparation = 18 h 1 report * 12 hours = 12 h total: 162.5 Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	TUDelft

NB2220	Statistical Physics	3
Responsible Instructor	Dr. J.W. Zwanikken	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course will give an introduction to statistical physics, covering the basic concepts of: emergent phenomena, the random walk, microstates and macrostates, the ideal gas, the ideal polymer, two definitions of entropy, temperature, thermodynamic potentials and ensembles, the Boltzmann distribution, the partition function, phase transitions, elecrostatic screening, and the Ising model. Knowledge will be applied to predict the elastic properties of ideal polymers, the barometric height distribution, the distribution of molecules in a potential landscape (in particular, an optical tweezer), the critical temperature of a fluid, and the coexisting densities of a bi-stable phase (with a focus on liquid-liquid phase separation). Ultimately, the course will reveal how the properties of materials and (living) systems are related to their components, and will offer methods to predict these properties. Expected prior knowledge: NB2011 Thermodynamics and Transport	
Study Goals	Learning objectives (LO) 1. Understanding the topics listed under `Course Contents', and how they are conceptually connected 2. Being able to analytically solve both elementary and advanced problems using the standard procedures of the theory. 3. Apply the concepts to get a deeper conceptual understanding of (living) materials. 4. Write clear and detailed solutions to problems, such that they can be understood by your peers.	
Education Method	Lectures: LO1, LO2, LO3 Homework: LO1, LO2, LO3 Exercise classes: LO1, LO2, LO3 Peer grading: LO4	
Assessment	To pass you always need to score at least 5.8 on the final exam. The final grade will be based on the final exam, plus other elements if they have a favorable effect on the final grade. Their relative weight is expressed in terms of points: Homework: The best 7 out of 8 HW sets will be counted if they increase the overall grade, with a maximum of 50 points Attendance: Will be counted if it increases the overall grade, with a maximum of 20 points Group assignments (2): Will be counted if they increase the overall grade, with a maximum of 30 points Midterm exam: Will be counted if it increases the overall grade, with a maximum of 100 points Final Exam: 300 points @Retakers old homework will not count towards the final grade.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Studyload/Week	There will be 4 hours of lecture and 4 hours of work session per week, which should cover the entire study load per week. Preparation for the exams will take extra time, roughly 8 hours each + error bar. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	TU Delft	

NB2230	Biomolecular Structure and Function	3
Responsible Instructor	Dr.ir. J.H.G. Lebbink	
Instructor	Dr. A. Jakobi	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2A	
Course Language	English	
Expected prior knowledge	NB1102 Chemistry 1; NB1012 Biochemistry; NB1016 Molecular biology	
Course Contents	This course teaches the basic determinants and elements of biomolecular structure and dynamics, structure-function relations, and current methods in structure determination, with a focus on proteins and their complexes with other biomolecules. It furthermore provides practical information on how to navigate the publicly available structural databases and how to interpret, analyze and optimally visualize biomolecular structures.	
Study Goals	At the completion of this course the student will be able to: 1. Describe energetic principles, composition and elements that define biomolecular structure 2. Describe relations between structure, dynamics, interactions and function 3. Have insight into current methods for macromolecular structure determination 4. Be able to navigate structural databases 5. Be able to critically analyze biomolecular structures 6. Be able to create high-quality figures containing detailed biomolecular structure information	
Education Method	The course will be taught using lectures and guest lectures, computer assignments and homework to be handed in and discussed in class. Professional skills that will also be developed are reading of scientific papers, integrating and applying scientific knowledge, problem solving and time management.	
Computer Use	yes	
Literature and Study Materials	Recommended Book: The Molecules of Life by John Kuriyan ISBN: 9780815341888	
Assessment	Class and Homework Assignments (20%), Midterm Digital Exam (25%), Final Digital exam (55%).	
Enrolment / Application	For course and assessment of class and homework: Brightspace. For exam: Osiris Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The teacher can be reached at j.lebbink@erasmusmc.nl	
Studyload/Week	3ECTS = 84 hours = 8.4 hours/week Lectures and practicals 4 hrs/week * 8 weeks = 32 hours Homework 2 hrs/week * 8 weeks = 16 hours Exam (midterm and final) 2*3 hours = 6 hours Study course material at home 3hrs/week * 10 weeks = 30 hours Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Erasmus MC	

NB2240	Imaging 2	3.5
Responsible Instructor	Dr.ir. D. Brinks	
Responsible Instructor	Dr. H.J. Geertsema	
Module Manager	Prof.dr. A.B. Houtsmuller	
Module Manager	Dr. J.A. Slotman	
Instructor	Q. Tao	
Instructor	A.B. Wong	
Practical Coordinator	Dr.ir. G.J. Kremers	
Practical Coordinator	Dr. M.E. van Royen	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	This course is a continuation of the imaging learning line and assumes that imaging 1 has been followed. This course furthermore builds on previous courses: Analysis and Linear Algebra (Year 1), Physics 1 (year 1), physics 2 (year 2), Signals and Systems (year 2) and Computational Science (Year 2).	
Course Contents	Microscopy images contain a wealth of information. Extracting this information in a way that is not subject to error requires 1) applying of the physics of image formation to assess the limits of physical information present; 2) applying of the mathematical concepts underlying digital image formation and analysis to allow for objective and complete information extraction; 3) application of the concepts of different microscopy methods to assess what method should be used to obtain a specific type of information and 4) the ability to automate image analysis tasks by using a computer and appropriate methods to efficiently, accurately and reproducibly extract objective, complete and physically relevant information. In this course the student will apply the knowledge obtained in imaging 1 in practice. The student will learn to analyse complex imaging data from relevant papers, and perform experiments involving advanced microscopy concepts. The student will be introduced to advanced imaging and image analysis concepts, among others Single molecule localization Microscopy (SMLM), Total internal reflection (TIRF), Structured Illumination microscopy (SIM), fluorescence recovery after photobleaching (FRAP), tracking, registration, data visualization and neural networks. These advanced concepts, as well as the basic concepts from imaging 1, will be brought into practice as the student will be guided through the most important, up-to-date fluorescence microscopy techniques for quantitatively studying cellular function and dysfunction at the nanoscopic scale. The students will work (in groups of five) on various topics (four will be chosen from a total of six) using different advanced fluorescence bioimaging techniques: cell division (mitosis and meiosis) will be studied using confocal time lapse microscopy and single molecule localisation (SMLM) (topics 1, 2), the assembly of microtubules will be studied using total internal reflection (TIRF) microscopy and the role of focal adhesions in migration using structured illumination microscopy (SIM) (topics 3, 4), gene transcription regulation will be studied by fluorescence recovery after photobleaching (FRAP) (topic 5), and cell membranes will be visualised and cell volume/surface ratios will be determined using specific membrane and nuclear staining and 3-D confocal microscopy (topic 6). For each topic the students will use the various state-of-the-art research microscopes to record images themselves (under supervision), will analyse the data, will write a report and in a final session will discuss the reports with the supervisor in a 15 minute session. Topics not dealt with in the practical sessions will be dealt with in the review of relevant papers, where students will be asked to use the advanced concepts introduced to extract relevant info and discuss the paper and its methodologies. The course is closely related to all Nanobiology courses as a variety of cell biological model systems are investigated using microscopes, analysed using mathematical techniques and reported in writing and discussed afterwards	
Study Goals	A. Demonstrate understanding of the basic principles of 1. image registration 2. tracking and motion analysis 3. volume and surface rendering 4. machine learning techniques in image analysis B. demonstrate understanding of the basic principles of: 1. fluorescence and green fluorescent protein (GFP)-tagging, and fluorescence stainings 2. wide field microscopy, confocal microscopy, TIRF microscopy, structured illumination microscopy and single molecule localization microscopy 3. quantification of fluorescent signals in microscope images, 4. fluorescence recovery after photobleaching (FRAP) experiments C. write well-structured reports, including discussion, on a (set of) microscopic experiment(s), D. demonstrate understanding of the content of the reports in a 15 minutes session with a supervisor. E. Demonstrate the ability to extract image-based information from scientific papers and present this information. F. Demonstrate the ability to write an advanced image analysis pipeline in imageJ.	
Education Method	Lectures, and tutorial sessions. The tutorial session will be microscopy practicals, computational practicals, or provide the ability to work on your assessments.	
Literature and Study Materials	Microscopy and Analysis: lecture videos, handouts of the lectures and computer exercises available on brightspace, as will be the papers that the students will read. Binders with information on the microscopes and assays will be available for the practical groups.	
Assessment	This course does not have a final exam. Students will be required to complete several assessments throughout the course, which will be graded for part of the final grade. These include: - Participation in lectures, practical sessions, and paper discussions - An advanced image analysis script based on one of the papers - A presentation of a paper - Group practical reports of the microscopy practicals - Discussion of the microscopy practical results with the supervisor	
Enrolment / Application	Enrolment in Brightspace is considered registration for any assessments, students must enrol in Brightspace for the course no later than the first week of classes.	
Studyload/Week	Imaging 2 is a 3.5 ECTS course divided over 10 weeks, of which 8 weeks are lecture and practical weeks. 3.5ECTS means a	

	time commitment of 98 hours. You will have 4 -6 contact hours per week for the first 8 weeks (you will have microscopy practical once every two weeks). This means about 5 hours of self-study is expected for the first 8 weeks, which should go to paper study, your FIJI script assignment, your practical reports and your paper reports; after which you have about 20 hours to prepare in groups for the final paper presentation and discussions with the supervisors. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	EMC

NB2330	Imaging 1	5
Responsible Instructor	Dr.ir. D. Brinks	
Responsible Instructor	Dr. H.J. Geertsema	
Instructor	Dr. F. Bociort	
Instructor	Dr.ir. D. Brinks	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	This course builds on previous courses: Analysis and Linear Algebra (Year 1), Physics 1 (year 1), physics 2 (year 2), Signals and Systems (year 2) and Computational Science (Year 2).	
Course Contents	Microscopy images contain a wealth of information. Extracting this information in a way that is not subject to error requires 1) understanding of the physics of image formation to assess the limits of physical information present; 2) understanding of the mathematical concepts underlying digital image formation and analysis to allow for objective and complete information extraction; 3) understanding of the concepts of different microscopy methods to assess what method should be used to obtain a specific type of information and 4) the ability to automate image analysis tasks by using a computer and appropriate methods to efficiently, accurately and reproducibly extract objective, complete and physically relevant information. In this course the student will learn the basic concepts, math and physics of imaging. The course starts with the basic physics of optical systems: - Electromagnetic waves, Polarization - Interference, Diffraction, Resolution limit - Geometrical optics, Imaging, Paraxial approximation, Thin lenses - Visual optical systems, Simple microscope, Matrix formalism for imaging The students will then learn about some basic types of microscopy utilizing the physical concepts learned before: - Bright field and dark field microscopy and phase contrast microscopy - Polarisation microscopy - Fluorescence microscopy and Confocal scanning microscopy Finally, the student will learn how to quantitatively analyze and extract basic information from the images made by types of microscopy introduced by computer, and will develop the skills to use computational image analysis tools for biological research.	
Study Goals	1. The student will have a good understanding of the basics of optics, including imaging and the knowledge necessary for understanding the construction of a traditional microscope. 2. The student will be able to apply the theory to solve simple problems. 3. The student will have a good understanding of several microscope types, including the general optical design and the interpretation of images. The types of microscopes are: bright and dark field microscopes, phase contrast microscopes, polarisation microscopes, fluorescence microscopes and confocal laser scanning microscopes. 4. The student will understand the physical, mathematical and computational limits of information contained in images 5. The student will be able to process (filter/enhance) and analyze microscopy images. 6. The student will be able to select the appropriate image segmentation approach, segment relevant features and compute quantitative properties and measures of these features 7. The student will be able to use relevant image processing and analysis tools within ImageJ and write processing and analysis pipelines as scripts.	
Education Method	Lectures, lecture videos and exercises. Lectures will either be full lectures or flipped classroom activities;depending on the topic the lecturer may lecture in class or flip the classroom and request the lecture videos to be watched for a more in depth activity in class. Exercises consist of mathematical and computational tasks aimed at deepening the understanding of the physics of imaging and the mathematics of image analysis The content of the lectures and lecture videos is subject to change, partly because student feedback from the 2023-2024 cohort will only be available after this study content needs to be finished. However, there will always be opportunity to ask questions and obtain clarification, and practice exercises in class during the exercise hours. The exercise hours will be supported by TAs.	
Literature and Study Materials	Optics: For all lectures there will be hand-outs of the lecture material on Brightspace. The hand-outs and the exercises (with solutions) are in principle sufficient for preparing the exam. Students who are interested in a deeper understanding of the topics can read E. Hecht, Optics, any edition Microscopy: The student will have lecture notes and exercises available on Brightspace but some chapters of the book below will be required for further reading: D. B. Murphy, Fundamentals of Light Microscopy and Electronic Images, J. Wiley&Sons. This book can be downloaded from the library (per chapter). The exam will be based on the lecture notes, exercises and the specific chapters of the book. Analysis: lecture videos, handouts of the lectures and computer exercises available on brighspace are sufficient to prepare for the exam	
Assessment	Written examination, both for the main exam and for the resit.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrolment in Brightspace is considered registration for any and all other assessments, students must enrol in Brightspace for the course no later than the first week of classes.	
Studyload/Week	Imaging 1 is a 5 ECTS course divided over 10 weeks, of which 8 weeks are lecture and exercise weeks. 5ECTS means a time commitment of 140 hours. You will have 6 contact hours per week for the first 8 weeks. This means 8 hours of self-study is expected for the first 8 weeks, after which you have 28 hours for exam preparation in the final 2 weeks. Self-study can consist of reviewing the material or exercises or watching the lecture videos in preparation for a flipped classroom session. If a flipped classroom session will happen, this will be communicated. The easiest way to go through the course is to keep up with the lectures, lecture videos and practical exercises. The contact sessions allow you to ask for clarification about the material found in that week's lectures and videos and are most valuable if you have followed the prior lectures. The exercises allow you to work on the concepts of the week with the help of the TAs, and are most valuable if you have in fact watched the concepts of the week in the lecture videos, or attended the lectures, and are working on the practical of the week so that the TAs and lecturers can help you in the best way. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for	

	you
Location	Delft

TN2545	**Systems and Signals**	**6**
Responsible Instructor	Dr. F.M. Vos	
Instructor	Dr.ir. D.J. Verschuur	
Contact Hours / Week x/x/x/x	3x 2 uur p/w hoorcollege 3x 2 uur p/w werkcollege	
Education Period	2A 2B	
Start Education	2A	
Exam Period	2A 2B 3A	
Course Language	English	
Expected prior knowledge	Complex numbers, elementary calculus, including derivatives & integrals, solving linear equations with few unknowns, geometric series, trigonometric & exponential functions. Create and analyze graphs of elementary functions	
Course Contents	This course covers the following topics: Systems and signals in continuous and discrete time; Basic deterministic signals; Convolution and properties of LTI systems; Fourier series and Fourier transforms with their properties; Filters in theory and practice: first- and second-order DE, Butterworth filters, Gaussian derivative filters; the estimation of spectra in a time window and the associated uncertainty relation; the DFT and FFT as practical tools; introduction to 2D Fourier transforms; modulation and demodulation techniques; the analytic signal and Hilbert transforms; sampling (Nyquist theorem), aliasing and reconstruction; laplace and Z-transforms with their ROC and properties; feedback systems and associated stability determination.	
Study Goals	The main learning goals of this course on Systems and Signals are to enable that a student can: 1. recognize and determine basic properties of systems including linearity, time invariance, impulse response, and determine their response based on convolution. 2. derive Fourier series coefficients and Fourier transform of elementary signals. 3. describe the effect of frequency-dependent filtering of discrete and continuous-time signals and relate this to linear difference/differential equations with constant coefficients. 4. analyze the effect of sampling of band-limited signals and reconstruct a continuous-time signal from the discrete samples. 5. perform amplitude-modulation to elementary signals and derive and analyze associated reconstruction methods. 6. assess properties of systems based on the Laplace and Z-transform.	
Education Method	Oral lectures by teachers, practical working classes with supervision	
Literature and Study Materials	A.V. Oppenheim and A.S.Willsky, with S.H.Nawab, Signals and Systems, Prentice-Hall, 2nd edition, Ch. 1 to 11 (obtainable in bookshop and VVTP). Reader with additional material en lecture notes/hand-outs.	
Assessment	There are 3 bonus tests during the teaching period before the exam takes place. They are optional and can give a student a bonus to the final grade (10% contribution per bonus test). If a bonus test grade (BG) is lower than the exam grade (EG) it will not contribute. Therefore, the final grade (FG) is computed as one of these options, depending on the bonus grades: FG = 0.7 EG + 0.1 (BG1+BG2+BG3) FG = 0.8 EG + 0.1 (BG1+BG2) FG = 0.9 EG + 0.1 BG1 FG = 1.0 EG Bonus test results are only valid in the academic year in which they are obtained.	

Year	**2024/2025**
Organization	**Applied Sciences**
Education	**Bachelor Nanobiology**

3e jaar BSc NB 2024	
Responsible Program Employee	J.C. Colgrove
Contact for Students	Students with questions about minors or electives should contact the Academic Counselor <Studieadviseur-NB@tudelft.nl>
Introduction 1	The 3rd year of the programme consists of 3 key parts. * A one semester long minor, 30 EC * The Bachelor End Project (Thesis), 20 EC * Nanobiology Electives, 10 EC The minor is scheduled in the first semester of the third year. Nanobiology students may freely choose any available 30 EC minor offered by TU Delft, Erasmus MC*, or Leiden University. With approval from the Board of Examiners they may also select a minor at a different university or a self-composed minor (including minors abroad). *Erasmus MC minors are typically 15 EC, which means approval falls under self-composed minors because it's combined with other courses.

NB3000	Bachelor Thesis Project	20

Responsible Instructor	Dr. J. Pothof
Responsible Instructor	J.C. Colgrove
Course Coordinator	L. van der Elst
Course Coordinator	Ir. K. Immerzeel
Contact Hours / Week x/x/x/x	self study, approximately 28 hours per week.
Education Period	3A 3B 4A 4B
Start Education	3A
Exam Period	Exam by appointment
Course Language	English
Course Contents	The Bachelor End Project (BEP) is the last part of your Bachelor of Science in Nanobiology. In this project the knowledge and skills learned in the first two years is applied on a level that enables you to successfully follow a connecting Master programme
Study Goals	Student is able to 1. Understand and directly apply theoretical knowledge to research 2. use scientific methods and approaches in the areas of responsibility, communication, literature study, critical attitude and time planning. 3. Do research work including developing significant original ideas, and improving experimental setups including for safety. 4. Write an accurate independent report of a quality that it can be used for further research. 5. Present a clear, well-argued defense of their project and able to answer questions about it. 6. Be an effective team member with good English writing skills. Student shows open mindedness and creativity. The full grading scheme can be found at the TNW Eindprojecten Brightspace page.
Education Method	Individual project in a research group at TUDelft or ErasmusMC. Activities include independent research, lab meetings, writing, presenting, data analysis.
Assessment	- A report (see document BEP thesis guidelines in Brightspace page for Thesisoffice TNW) - A final presentation usually 20-30 min open to the public. - A defense. A 30-45 minute closed session with student and assessment committee where student can defend their thesis. This committee will consist of at least two examiners that are member of scientific staff (tenured or tenure track). 1. The responsible supervisor is one of the examiners. 2. Both examiners must be on the Nanobiology GreenList, one must be a teacher in the Nanobiology bachelor or master programme as indicated on the GreenList. The teaching member of the assessment committee must be identified at the time of application submission. The GreenList is available from the TNW EindProjecten BrightSpage page.
Enrolment / Application	To start this project the student must have completed all year 1 courses and at least 60 EC from year 2 and 3. The BEP enrollment form must be submitted and accepted by the Thesis office before beginning the project. Students who are doing a project in an Erasmus MC research group, must complete the necessary access paperwork before they can begin their project. Please check the Nanobiology Bachelor Brightspace page for instructions. This should be submitted at least 3 weeks prior to beginning the project.
Special Information	Students may not do a project with the same supervisor or in the same research group for a BEP and an Independent Research Project or Honours Research Project.
Remarks	Check on the Brightspace page of the endproject administation / Thesis office TNW for detailed information and required forms. 1. A Bachelor End Project earns the student 20 ECTS and is usually completed in one semester consisting of 20 weeks. 2. This is planned for semester 2 of year 3 (from Second week in February to First week in July usually), but can be started at a different time depending on study progress. 3. To start this project the student must have completed all yr 1 courses and at least 60 points from year 2 and 3. 4. Students are responsible for finding an available supervisor. The supervisor must be on the Nanobiology "GreenList" which is available on the Thesis Office BrightSpace page. Together the student and supervisor decide on a project. The student then completes and submits the BEP Approval form before they begin work on the project. This is checked by the programme coordinator or study advisor before the student is allowed to start. 5. The projects should be research work within an existing project in a lab. They should have substantial practical and theoretical components and lay the foundation for eventual independence. 6. The project is completed by a written report (BSc) thesis and oral evaluation procedure. 7. The student is expected to take part in research work about 28 hours per week. This includes participating in lab meetings, journal clubs and the like, as scheduling permits, while also completing their elective courses. Note that the end grade will only be registered when your (digital) end-report is handed in to the Thesis office. You will automatically receive a digital survey. Info via: ThesisOffice-TNW@tudelft.nl Thesis Office is located at the TN building, room A208

Year	**2024/2025**
Organization	**Applied Sciences**
Education	**Bachelor Nanobiology**

3e jaar BSc NB keuzevakken 2024	
Responsible Program Employee	J.C. Colgrove
Introduction 1	Students must take at least 10 EC of electives. Please see the study guides for each course for information about enrollment requirements. Some electives have limited enrollment. Instructions on how to enroll are included on the study guide for those electives. Students may request to use other courses as electives for example as part of a bridging programme. The request must be made and approved before beginning the course. This is done using the Change Courses form available on the standard forms page. https://www.tudelft.nl/en/student/faculties/as-student-portal/education/forms

NB3014	Computational Neuroscience	2.5
Responsible Instructor	Ir. M.J.F. Negrello	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/4/0	
Education Period	4 4A	
Start Education	4 4A	
Exam Period	4 4A 4B	
Course Language	English	
Course Contents	Biologically inspired neural networks perform a range of computational functions analogous to cognitive processing in the living brain, e.g., as pattern recognition, associative learning, motor control, forecast and prediction. In this course we will describe basic principles and algorithms underlying these functions. With a sandbox neural-network construction software we will implement these networks and familiarize with their properties and functions.	
Study Goals	At the completion of this course the student will be able to: 1. Produce biophysically accurate neuronal models. 2. Interpret neuronal activity with dynamical systems theory. 3. Produce spiking networks. 4. Explain how neurons may recognize input patterns through the perceptron analogy 5. Train a neural network to perform pattern recognition. 6. Intuitively understand how recurrent connections of a recurrent neural network leads to complex dynamics	
Education Method	Computational projects (python) and Lectures Individual/group programming assignments Skills acquired in this course: - Reading and writing python code (from the python projects) - Obtaining and presenting new knowledge (acquired in the final project) - Giving reviews (on peer's projects) - Writing report - Presenting - Team work	
Books	Educational material provided as jupyter notebooks. Selected articles will be provided.	
Prerequisites	- biology of the cell - biophysics of cell membranes (membrane potential, concentration) - python (numpy, matplotlib) - linear algebra (dot and cross products, eigenvectors and eigenvalues) - calculus (differential equations)	
Assessment	Students will be peer-evaluated via a project presentation (50%) and a report (50%) Projects will be groups of 1 or 2. Reports and presentations are given by the group. Students get one joint grade for the presentation and an individual grade for your report. Students pass if the combined grade meets the minimum. RESIT: - Deliver own code implementing a neuronal model and a 3000 word report about its limitations/improvements/assumptions.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The teacher can be reached at m.negrello@erasmusmc.nl	
Elective	Yes	
Tags	Abstract Artificial intelligence Lineair Algebra Mathematics Modelling Numeric Methods	
Studyload/Week	4h/week * 5 weeks = 20 h (lectures and working classes) ca. 8h preparation of final presentation and report. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.	
Location	Erasmus MC Rotterdam	

NB3015	A primer in Neuroscience	2.5
Responsible Instructor	Ir. M.J.F. Negrello	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/0/4	
Education Period	4 4B	
Start Education	4B	
Exam Period	4 5 4B	
Course Language	English	

Course Contents

NB3015 is a flip-the-classroom experience, with emphasis on developing the ability to read and grasp essentials from neuroscience topics, while emphasizing a mechanistic understanding of brain function. We will be reading chapters from "Principles of Neuroscience" (5th Edition) and students will present them in a pre-established sequence

Week 1. Introduction and Plan of Attack (Lecture)

Week 2. Sensation and Perception (Student Presentations)

Week 3. Sensorimotor Behavior (Student Presentations)

Week 4. Cognition (Student Presentations)

Presentation Rules

- Presentations will last for 20'+5-10' (Q&A) and will involve groups of maximum 2 people. Note: Depending on the number of students taking the course the presentation time allotted may vary.

PS. Though not mandatory, NB3014 (Computational Neuroscience) is a helpful pre-requisite.

Study Goals

Learning Goals

1. Neurons and Synapses
1.a Know different kinds of cortical neurons and their distinguishing features.
1.b Know the principal (structural and functional) differences between excitatory and inhibitory synapses.
1.c Know the principal neurotransmitters of the brain.
1.d Give examples of processing hierarchies in the nervous system.
1.e Describe connectivity patterns between neurons in the cortex.

2. Sensation and Perception
2.a State the goal of the study of psychophysics.
2.b Be able to explain what are psychophysical laws.
2.c Describe probabilistic strategies to quantify perception.
2.d Give examples of sensory features encoded by neurons.
2.e Know the role of the somatosensory systems in the control of behavior.

3. Perception and Movement
3.a Describe the central problem of cognitive science.
3.b Explain what is a neural map and how it is obtained.
3.c Know basic differences between the sensory and motor somatotopicmaps.
3.d Be able to describe what is an internal model.
3.e Be able to differentiate between forward and inverse models.
3.f Be able to explain what is segmentation.
3.g Understand the hierarchical organization of visual processing.

4. The Organization of Cognition
4.a Be able to produce a definition of cognition.
4.b Be able to give examples of serial and parallel processes in the brain.
4.c Be able to explain why receptive fields size grows during information processing by the brain.
4.d Be able to explain what are association cortices, and name examples.
4.e Understand the inextricable relationship between action and perception in the sensorimotor loop.
4.f Be able to describe the different kinds of memory.
4.g Be able to name the molecular mechanism responsible for long term memory storage.
4.h Know the differences between imaging methods in terms of goals
4.i Explain how neural correlates of consciousness be measured with imaging.

Education Method

Students read and summarize selected chapters of "Principles of Neural Science" according to specific learning goals.

Skills:

- Reviewing, summarizing and presenting complex knowledge
- Receiving and giving peer feedback
- Writing a short scientific report

Books

Kandel - Principles of Neuroscience
Gerald Edelman - Topobiology
Ross Ashby - Design for a Brain
Varela and Maturana - The tree of knowledge
Llinas - I of the Vortex

Prerequisites

NB3014 (recommended)

Assessment

Evaluation Rules (NB3015):
The evaluation is based on 2 components (50/50). Students pass if they achieve the combined passing grade.

1) The presentation grade:

1.1 The grade of a student presentation where 30% of the grade is from 'peer-evaluation', and 70% is due to lecturer evaluation.

1.2 The evaluation of presentations (peer and lecturer) is done according to four dimensions: Form (slides), Content (selection of material), Clarity (of explanation), Q&A.

	1.3 Grade for presentation is shared equally by the two presenters 2) An essay: 2.1 The essay focuses on a different topic than your presentation. From all of the other presentations, you select a topic that has piqued your interest and expound on it. The essay can take the form of a critique, a different question, an observation, a proposed hypothesis, an experimental question. 2.2 Suggested word count: 2000. 2.3 Grade will depend on correctness, clarity and concision. RESIT: A 5000 word essay combining two of the possible presentation topics.
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.
Special Information	The teacher can be reached at m.negrello@erasmusmc.nl
Studyload/Week	In class we have 4h per encounter, for a total of 5 encounters, thus 20h. Extra class hours amount for about 4h for the group that is preparing presentations. Reviewing the week's material should take about 4h per week. Writing the final report should take about 4h. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you.
Location	Erasmus MC Rotterdam

NB3016	High Speed Scientific Computing	2.5
Responsible Instructor	Dr.ir. C. Strydis	
Responsible for assignments	Dr.ir. C. Strydis	
Contact Hours / Week x/x/x/x	0/0/0/0/8/0/0/0/	
Education Period	3 3A	
Start Education	3A	
Exam Period	Exam by appointment 3A 3B	
Course Language	English	
Expected prior knowledge	Python programming	
Course Contents	Scientific computing is a field of science making use of computers to analyze and solve scientific problems. It involves construction of mathematical models which permit quantitative analysis of complex phenomena. While an excellent tool for analysis, scientific computing comes at a price: Model simulations are often complex, resulting in slow simulations speeds. By considering various real-world scientific problems, this course will teach students hands-on the basics of algorithmic complexity, the tricks behind constructing efficient and parallelizable algorithms, and will introduce them to various computer technologies for executing high-speed simulations, which are orders of magnitude faster than existing approaches.	
Study Goals	At the completion of this course the student will be able to: 1. Understand the basics of scientific computing. 2. Encode simple algorithms into a high-level computer language (Python). 3. Understand the fundamentals of parallelism at different levels (data, instructions, processors etc.) and its importance for high-speed computing. 4. Re-encode simple algorithms into parallelized, accelerated versions (also in Python). 5. Master the parallel-processing and GPU-coding capabilities of Python. 6. Encode new and optimize existing computational problems, effectively reducing their simulation times from hours to seconds.	
Education Method	- Lectures (including live coding) - Computer-lab assignments in the Cloud - Lab reports and final exam	
Computer Use	Lab work will be performed on cutting-edge computers in the Cloud. Remote access to Cloud computers will be arranged annually as necessary to offer maximal lab flexibility but also to familiarize students with the Cloud-computing paradigm. Regarding programming languages, Python knowledge is required for use in the labs.	
Books	Lecture notes and supplementary materials will be provided during the lectures.	
Prerequisites	Python	
Assessment	Grading scheme: Pass/Fail. Lab reports are based on four lab assignments and contribute, in total, to 50% of the final mark. Lab assignments must be handed in on time, are graded independently and are compulsory to pass (as a whole). The final exam is a randomized multiple-choice test directly on Brightspace. It contributes to the remaining 50% of the final mark. The exam is graded independently and is compulsory to pass. The final mark is pass/fail and is calculated as the average of the lab-assignment mark and the final-exam mark. A student needs a final mark >=5.8/10.0 to pass, as per standard TUDelft regulations. Exam resit: This is only possible for students who failed to acquire a passing mark in the final exam and/or (in rare cases) a passing mark in the lab-assignment mark. In such cases, an oral exam is, most commonly, arranged in the weeks after the final exam. This has to be arranged with the course instructor based on time availability.	
Permitted Materials during Tests	No	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the last week before classes start.	
Special Information	The teacher can be reached at c.strydis@erasmusmc.nl and at c.strydis@tudelft.nl.	
Elective	Yes	
Tags	Algorithmics Computer Engineering Modelling Optimalisation Practicals Programming Programming concepts Technology	
Studyload/Week	4h/week lecture/instruction, 4h/week of practical time available with TAs plus 4-12 h/week at home. Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder, requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take you.	
Location	Lectures: Delft Campus. Labs: Delft Campus and via remote access.	

NB3017	Quantum mechanics for Nanobiology 1	2.5
Responsible Instructor	Dr. T. Idema	
Contact Hours / Week x/x/x/x	0/0/0/0/8/0/0/0	
Education Period	3 3A	
Start Education	3 3A	
Exam Period	3 3B 4A	
Course Language	English	
Required for	NB3018	
Expected prior knowledge	Physics 1A, 1B and 2, Analysis 1-3, Linear algebra, and Differential equations. The course will be fairly math-heavy, so a good understanding of the prerequisite math courses is essential.	
Course Contents	This description applies to both part 1 (NB3017) and part 2 (NB3018) of Quantum mechanics for Nanobiology: In this course we introduce the concepts and applications of quantum mechanics in the context of nanobiology. We will cover basic quantum ideas (particle-wave duality, atomic orbitals and shells, tunneling) and put them on a solid mathematical foundation (part 1). Building on this foundation, we also explore atomic emission and absorption spectra, the fundamental principle behind lasers, the quantum mechanics behind chemistry, and properties of solids, especially conductors, semiconductors and insulators (part 2). The cornerstone of the course (and quantum mechanics) is the Schrödinger equation, together with its solutions (known as the wavefunctions) and the associated statistical interpretation. As the Schrödinger equation is a differential equation, and the wavefunctions are its eigenfunctions, this course combines what students have learned in their earlier physics and math courses. In explaining the periodic table and the mechanism behind photosynthesis, it also directly connects to the courses chemistry and physical biology of the cell.	
Study Goals	At the completion of this course, the students will be able to: Indicate the difference between quantum and classical dynamics. Explain the concept of the quantum wave function and quantum operators, and perform mathematical operations with them. Solve the one-dimensional Schrödinger equations for a number of common potentials, including square and harmonic wells and barriers. Explain the idea behind the Heisenberg uncertainty principle, and apply it to arbitrary operators. Solve the three-dimensional Schrödinger equations for the hydrogen atom. Calculate emission spectra for atoms. Calculate angular momenta in quantum mechanics.	
Education Method	Lectures and tutorials. Lectures will be on-campus, material to be covered will be indicated on Brightspace, with references to the notes and the book. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.	
Books	D. J. Griffiths and D. F. Schroeter, Introduction to quantum mechanics, third edition (2018), Cambridge University Press, ISBN 978-1-107-18963-8.	
Assessment	Homework and final written exam; optional midterm. The final written exam will be combined with (and given at the same time as) that of NB3018. The final grade will consist for 25% of the homework and 75% of the exam grade; students can earn a bonus point for the exam with the optional midterm. Students taking NB3017 only (not NB3018) will take a modified final exam at the same time as students taking both courses. The grade will be calculated as homework+exam+bonus. Students should register for the NB3017 exam in Osiris. Students taking both NB3017 and NB3018 will take one final exam. The grade for NB3017 will be calculated as exam+bonus+NB3017 homework. The grade for NB3018 will be calculated as exam+bonus+NB3018 homework so the two grades may be different. Students should register for both the NB3017 and the NB3018 exam in Osiris. The retake will follow the same pattern as the regular exam. If there are very few participants, the retake may be an oral instead of a written exam.	
Remarks	The course requires students to actively participate throughout, and work seriously during all sessions. In principle, students should attend all sessions.	
Studyload/Week	Preparation 2x1 hour, lecture 2x2 hours, tutorial 2x2 hours, finishing homework 1x2 hours = 12 hours per week. Please note: every student is different and studies at a different pace. You may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you. Note that I've scheduled time for preparation; it will help to look at the material beforehand, so you have some idea of what is to come. Of the 70 hours scheduled for the course (2.5 ects), the 12 hours/week cover 48, leaving you ample room for additional self-study and preparation for the exam.	
Continuing Courses	NB3018	
Location	Delft	

NB3018	Quantum mechanics for Nanobiology 2	2.5

Responsible Instructor	Dr. T. Idema
Contact Hours / Week x/x/x/x	0/0/0/0/0/8/0/0
Education Period	3 3B
Start Education	3B
Exam Period	3 4 3B 4A
Course Language	English
Expected prior knowledge	Physics 1 and 2, Analysis 1-3, Linear algebra, and Differential equations. The course will be fairly math-heavy, so a good understanding of the prerequisite math courses is essential. This course immediately follows NB3017 (which is thus a key prerequisite). Obviously this part follows the course Quantum mechanics for Nanobiology 1 (NB3017).
Course Contents	This description applies to both part 1 (NB3017) and part 2 (NB3018) of Quantum mechanics for Nanobiology: In this course we introduce the concepts and applications of quantum mechanics in the context of nanobiology. We will cover basic quantum ideas (particle-wave duality, atomic orbitals and shells, tunneling) and put them on a solid mathematical foundation (part 1). Building on this foundation, we also explore atomic emission and absorption spectra, the fundamental principle behind lasers, the quantum mechanics behind chemistry, and properties of solids, especially conductors, semiconductors and insulators (part 2). The cornerstone of the course (and quantum mechanics) is the Schrödinger equation, together with its solutions (known as the wavefunctions) and the associated statistical interpretation. As the Schrödinger equation is a differential equation, and the wavefunctions are its eigenfunctions, this course combines what students have learned in their earlier physics and math courses. In explaining the periodic table and the mechanism behind photosynthesis, it also directly connects to the courses chemistry and physical biology of the cell.
Study Goals	At the completion of this course, the student will be able to: Explain the concept of spin, identify fermionic and bosonic particles, and apply the Pauli exclusion principle. Explain the shape of the periodic table. Use perturbation theory and the variational principle to study the properties of quantum mechanical systems without known analytical solutions. Explain the properties of conductors, insulators and semi-conductors with quantum mechanics. Explain how molecular bonds form based on quantum mechanics, and use the LCAO (linear combination of atomic orbitals) to approximate the ground state energy of simple molecules.
Education Method	Lectures and tutorials. Lectures will be on-campus, material to be covered will be indicated on Brightspace, with references to the notes and the book. Students work on problems themselves (with the help of TAs) during tutorials; some of these worked solutions are to be handed in each week and will be graded.
Books	D. J. Griffiths and D. F. Schroeter, Introduction to quantum mechanics, third edition (2018), Cambridge University Press, ISBN 978-1-107-18963-8.
Prerequisites	NB3017
Assessment	Homework and final written exam; optional midterm (after NB3017). The final written exam will be combined with (and given at the same time as) that of NB3017. The final grade will consist for 25% of the homework and 75% of the exam grade; students can earn a bonus point for the exam with the optional midterm. Students taking NB3017 only (not NB3018) will take a modified final exam at the same time as students taking both courses. The grade will be calculated as homework+exam+bonus. Students should register for the NB3017 exam in Osiris. Students taking both NB3017 and NB3018 will take one final exam. The grade for NB3017 will be calculated as exam+bonus+NB3017 homework. The grade for NB3018 will be calculated as exam+bonus+NB3018 homework so the two grades may be different. Students should register for both the NB3017 and the NB3018 exam in Osiris. The retake will be organized the same way as the regular exam. If there are very few participants for the retake, it may be given as an oral instead of a written exam.
Remarks	The course requires students to actively participate throughout, and work seriously during all sessions. In principle, students should attend all sessions.
Studyload/Week	Preparation 2x1 hour, lecture 2x2 hours, tutorial 2x2 hours, finishing homework 1x2 hours = 12 hours per week. Please note: every student is different and studies at a different pace. You may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for you. Note that I've scheduled time for preparation; it will help to look at the material beforehand, so you have some idea of what is to come. Of the 70 hours scheduled for the course (2.5 ects), the 12 hours/week cover 48, leaving you ample room for additional self-study and preparation for the exam.
Location	Delft

NB3020	Genomics Technology in Breast Cancer Research	2.5
Responsible Instructor	Prof. J.W.M. Martens	
Contact Hours / Week x/x/x/x	0/0/0/0/4/0/0/0	
Education Period	3 3A	
Start Education	3 3A	
Exam Period	3 3A 3B	
Course Language	English	
Expected prior knowledge	Basic knowledge in oncology	
Course Contents	Context of the course Our increasing understanding of diseases like cancer has led to new drugs becoming available for clinical application. While the application of these new drugs leads to increased cure rates, their high cost and their potential side effects however also pose a pressure on health care cost for society. In the era of personalized medicine, we aim to provide drugs only to patients benefitting from them. Genomics techniques help us to better position new and current treatments, yielding biomarkers discriminating patients sensitive from refractory to treatment. Furthermore, the basic knowledge gained through genomics research allows for new treatments to be developed further improving patient care. Aims of the course This course highlights state-of-the-art technological developments in the fields of genomics and how these tools can be applied to discover new biomarkers and what we learn from their use with regard to the understanding of mechanisms of disease progression. Breast cancer, a common cancer in humans is used as an example. The course specifically aims to provide an understanding of normal mammary gland development and how breast cancer can arise in epithelial cells from the mammary gland due to hereditary as well as somatically acquired mutational programs. Furthermore, it is discussed how this knowledge can used to optimize patient care allowing for patient stratification involving both predictive and prognostic markers and for monitoring disease using innovative diagnostic tools. Teaching activities: The course will be given in the format of classical lectures and will cover an introduction of breast cancer pathology, physiology and molecular biology. Also the genetics of the disease will be reviewed. Different types of technological platforms for genomics research will be reviewed as well as to some details the bioinformatics and mathematics behind the applied data analysis tools. The course also includes a journal club in which the students will present and discuss literature related to the topics and learning objectives of this course. Overall, during the course, students will get a clearer understanding of how technology and the mathematics behind it can be applied to relevant issues in the field of cancer research and aid in understanding disease progression and therapy resistance. During the lectures, students understanding of the matter will be actively tested. During a journal club, students will read assigned literature and prepare a presentation and propose discussion points for their peers. This elective course is complementary to the elective course on Nanobiology in Cancer, but it is not compulsory to take the latter in advance. This course is essential to understand the mechanisms of disease progression which helps the develop new treatment on the one hand but also can help in keeping cancer treatment cost effective. Students are expected to read the background literature provided as well actively participating to the lectures by posing questions. Furthermore, a critical attitude is expected when selected scientific literature related to the topics of the course is being prepared for during the journal club. At any moment during the course I am available for questions by email and will reply within 24 hours. During the preparation of the presentation for the journal club I will be present on site.	
Study Goals	Learning objectives: 1. Describe the essential cell types of the normal mammary development as well as the cell type from which breast cancer can arise (understand) 2. State the difference between driver and passenger mutations (understand) 3. Differentiate between prognostic and predictive markers (analyse) 4. Differentiate between low, intermediate and high risk genes/alleles for getting breast cancer and methods to detect them most efficiently (analyse) 5. Describe the essential steps necessary for breast cancers to become lethal and methods to prevent this from happening (understand) 6. Describe the key features of the three dominant mutational program in breast cancer and their clinical significance (understand) 7. Differentiate between intra- and inter-tumor heterogeneity (analyse) 8. Describe methods to detect biomarkers in liquid biopsies (understand)	
Education Method	- Lectures - Quizzes (optional) - Journal club - Presentations	
Assessment	Graded course. Students will be evaluated based on presence at lectures (5%), presentation of assigned paper for the journal club (20%), and written examination and/or quizzes (75%).	
Enrolment / Application	This course is offered in 3A and has limited enrollment. Enrollment will be available after November 1 via the Nanobiology Bachelor Brightspace page and is done on a first come, first served basis with priority for Nanobiology students. Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The teacher can be reached at: j.martens@erasmusmc.nl	
Tags	Analysis Group work Lab Research Research Methods Software Technology	
Location	Erasmus MC, Rotterdam	

NB3021	Optics and its application in Nanobiology	2.5
Responsible Instructor	Dr. F. Bociort	
Instructor	Dr.ir. D. Brinks	
Contact Hours / Week x/x/x/x	0/0/0/0/0/4/0/0	
Education Period	3 3B	
Start Education	3 3B	
Exam Period	3 4 3B 4A	
Course Language	English	
Required for	For students who intend to follow the Applied Physics Master Programme, this course can replace the course TN2345 in the Bridging Programme	
Expected prior knowledge	The Optics part of this course is a continuation of the Optics part of the course NB2041 Optics and Microscopy.	
Course Contents	Maxwell's equations and electromagnetic waves in vacuum and non-conducting materials, electric dipole radiation, dispersion, Huygens-Fresnel Principle, Fermat's Principle,reflection, refraction and transmission of electromagnetic waves at an interface between two media, Fresnel equations for reflectance and transmittance, total internal reflection and evanescent waves, fiber optics, optical tweezers and magnetic tweezers to study forces in biology, fluorescence lifetime imaging microscopy, Förster resonant energy transfer, optogenetics controlling cells with light, multiphoton microscopy, introduction to nonlinear optics, optics of short-pulsed lasers, coherent anti-Stokes Raman scattering microscopy	
Study Goals	The student will have a good understanding of the basics of optics and several applications in nanobiology. The student will be able to answer questions and to apply the theory to solve simple problems.	
Education Method	lectures	
Literature and Study Materials	hand-outs of the lecture material optional: E. Hecht, Optics, any edition	
Assessment	Written examination, both for the main exam and for the resit. Normally the exam is closed-book, off-campus exams are open-book.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Location	Delft	

NB3022	Epigenetics	2.5
Responsible Instructor	Dr. H. Mira Bontenbal	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/0/4	
Education Period	4 4B	
Start Education	4 4B	
Exam Period	4 4A 4B	
Course Language	English	
Expected prior knowledge	Prior to the beginning of this course, students are expected to have knowledge of transcription/gene regulation, cell and developmental biology. Students should have completed course(s) Molecular Biology, Physical Biology of the Cell 1 and Evolutionary Developmental Biology. The course will assume knowledge of cellular biomolecules gained in the courses of Molecular Biology and Physical Biology of the Cell 1. Concepts of cell differentiation and development of Physical Biology of the Cell 1 and Evolutionary Developmental Biology are particularly important for this course. Knowledge from Molecular Biology about genes, transcription factors and chromatin are also required.	
Course Contents	How is it that our bodies have hundreds of different cell types, each having very different patterns of gene expression and well-defined behaviours and functions, and yet all have the same genome? Why do the cells in your skin divide into two other skin cells but not into two neurons or two muscle cells? How is it possible that our diets will affect our children and even grandchildren? All these facts have something in common which is the heritability of information across cell generations, without modifications to the DNA sequence. The field that studies this is Epigenetics. In this course, the students will deepen their understanding of the epigenetic mechanisms that mediate gene expression changes, development and disease. The course will be divided into three parts: 1. The first part will be devoted to describing what the different epigenetic mechanisms are, such as histone modifications and DNA methylation, that regulate genes in general, but also at the level of two paradigms in epigenetics, imprinted genes and X chromosome inactivation. We will also highlight the role of nuclear organization in gene expression control. 2. In the second part of this course, we will study how epigenetics affect development and disease, and we will look into epigenetic modulators that are used to combat disease. We will also discuss prions as a non-traditional epigenetic mechanism that can lead to disease. 3. Lastly, the students will learn what the different techniques are that are currently available to study epigenetic modifications and their implications, from basic biology to the medical field. This elective course is very relevant to Nanobiologists who want to expand their knowledge of epigenetic control of gene expression, how epigenetics is critical to understanding cell differentiation and development, and some types of diseases and cancer progression, and what the current technologies are used to study epigenetics in basic biology, medical research and biotechnological companies.	
Study Goals	At the completion of this course, the student will be able to: 1. Rephrase what epigenetics is. 2. Cite and explain the different types of epigenetic modifications that exist in mammals and what their mechanisms of action are in regulating gene expression. 3. Describe what imprinted genes and X chromosome inactivation are. 4. Explain how epigenetics can modulate disease and cancer progression. 5. Cite and explain the existing current technologies to study epigenetic mechanisms of gene expression control. 6. Rephrase essential concepts of epigenetics 7. Apply the obtained knowledge in order to critically describe, interpret and explain the results of scientific experiments in epigenetics, using provided text and figures from scientific manuscripts. 8. Read and critically assess scientific literature (soft skill). 9. Collaborate with fellow students to prepare a scientific presentation (soft skill). 10. Present a scientific argument, engage in discussion with fellow students and answer questions in public (soft skills) 11. Give peer feedback of presentations (soft skill)	
Education Method	- Interactive lectures - Interactive practicals - Homework (graded) - Workshop: analysis of genome-wide data - Group work: students will work in group to prepare a presentation and present in front of their peers. There will be peer feedback, which will count for your presentation grade. Soft skills developed in this course: group work, presentation, peer review of presentation.	
Books	No book is required for this course. However, most of the figures in the lectures and practicals come from Epigenetics 2nd Edition (Allis, Caparros, Jenuwein, Reinberg, Lachner). All chapters of this book are open-access published reviews. A repository of the figures and their legends can be found at http://learnepigenetics.com/	
Assessment	Assessment: 20% Homework Assignments 20% Oral group presentation 60% Exam. (they can compensate each other) If the exam and re-exam are not passed, the grades for the homework and presentation are kept for next year. There is no resit for the Presentation. An exceptional alternative homework assignment will be given in case the student cannot do the regular homework assignment.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Special Information	The teacher can be reached at: h.mirabontenbal@erasmusmc.nl	
Remarks	Hegias Mira Bontenbal can be reached at h.mirabontenbal@erasmusmc.nl	
Studyload/Week	2.5 ECTS 4h/week * 4 weeks = 16h in-class hours ca. 8h / week * 4 weeks = 32h study at home, homework, group work for presentation preparation 20h exam preparation Please note: every student is different and studies at a different pace. Each may find some topics easier and others harder requiring more time for mastery. This is a loose approximation and in no way guarantees this is how much work it will take for	

	you.
Schedule	I will put the up-to-date schedule on Brightspace. So please follow that schedule and not the schedule on mytimetable.tudelft.nl. Use the Brightspace one when the course starts.
Location	Erasmus MC, Rotterdam

NB3023	**Human Complex Genetics**	**2.5**
Responsible Instructor	Dr. J.G.J. van Rooij	
Instructor	L. Broer	
Contact Hours / Week x/x/x/x	0/0/0/0/0/4/0/0	
Education Period	3B	
Start Education	3B	
Exam Period	3B 4A	
Course Language	English	
Expected prior knowledge	Basic knowledge of human genetics is recommended but not required.	
Course Contents	This course provides an introduction into the field of complex genetics. Also referred to as population genetics; generating large-scale genetic datasets and then answering relevant biological or clinical questions. The course is split in four contact days and and exam: Day 1: Introduction to complex genetics (genetic variation, genome-wide association studies - GWAS, applications of complex genetics). Day 2: Interpretation of GWAS results (GWAS analysis methodology, biological interpretation of genetic variants). Day 3: Clinical application of GWAS results (disease risk prediction, polygenic risk scores, clinical decision-making). Day 4: Discussion of assignments; practical examples performed in groups or extracted from literature. Through these topics the course covers the basics of performing complex genetic studies, as well as some of the most common and currently utilized applications.	
Study Goals	At completion of this course, the student will be able to: 1. Indicate the difference between clinical and complex/population genetics 2. Perform a complex genetic analysis (genome-wide association study) 3. Interpret complex genetic results in biological pathways 4. Calculate polygenic risk scores and apply them to predict disease risk 5. Indicate the (possible) role of genetics in medicine	
Education Method	The course is hosted in part by researchers from the Human Genotyping Facility (HuGe-F) at Erasmus MC. It focuses on the practical aspects of complex genetic studies and provides you hands-on experience and practical examples. Each course day contains a lecture (~1h) followed by a practical session on the topic of that day (first three course days). The practical sessions will be held using pre-prepared scripts in R. The students are invited to work on a practical example in small groups and discuss this in the 4th day. The course will develop professional skills; data analytics, interpretation of results, group work, presenting results.	
Assessment	Homework and practical assignment to be discussed on day 4. Individual exam on day 5.	
Permitted Materials during Tests	Open book exam.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Remarks	To contact the teacher, please use the EMC email address: j.vanrooij@erasmusmc.nl.	
Studyload/Week	This course stands for 2.5 ECTS, or approximately 14h/week. Approximately 4h each week is reserved for the lecture and practical sessions. The remaining time is intended for homework (week 1-3), preparing the group assignment (week 4) and preparing for the exam (week 5).	
Location	Rotterdam	

NB3024	Advanced Math Topics	2.5
Responsible Instructor	S.M.F. Abdoel	
Instructor	D.M. Hoonhout	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/4/0	
Education Period	4 4A	
Start Education	4A	
Exam Period	4A 4B	
Course Language	English	
Expected prior knowledge	Prerequisite courses are Differential Equations (NB2191) and Computational Science (NB2181).	
Course Contents	Many physical phenomena can be described by differential equations. In the context of Nanobiology you can think of processes like heat dispersion, diffusion or chemical reactions. Often these equations have to be solved numerically. First the course will discuss how a system of ordinary differential equations (ODEs) can be solved using numerical time integration. Then we discuss how a stationary diffusion equation equation can be discretized with the Finite Difference Method. Finally we combine time integration and space discretization to numerically solve the heat equation. Lesson 1. Numerical integration and some time integration methods: Forward Euler, Backward Euler, Trapezoidal Method, Modified Euler. Implicit methods and Explicit methods. Predictor-corrector methods. Lesson 2. Stability of time integration methods, order of accuracy and systems of ordinary differential equations (ODEs). Lesson 3. Numerical differentiation and Finite Difference Method(s). Discretization of boundary conditions and properties of the discretization matrix. Lesson 4. Finite Difference Method (continued): (Non-constant) diffusion and the heat Equation. The method of lines.	
Study Goals	General: given an initial value problem or boundary value problem, the student can make a motivated choice of appropriate numerical method to find a numerical approximation of the solution, and analyse the quality of the approximation. More specifically: 1. The student knows, can apply and can implement numerical time integration methods to solve ordinary differential equations (ODEs) and initial value problems, such as Forward Euler, Backward Euler, Trapezoidal Method and Modified Euler. 2. The student can classify methods as implicit method, explicit method and/or predictor-corrector method, and can analyse strengths and weaknesses. 3. The student can analyse stability and order of accuracy of time integration methods for (systems of) ODE(s). 4. The student can apply, implement and analyse the discretisation of the 1D diffusion equation with the Finite Difference Method. 5. The student can numerically solve the time-dependent 1D heat equation with the method of lines.	
Education Method	Four lecture classes and four instruction classes with programming assignments.	
Literature and Study Materials	Numerical Methods for Ordinary Differential Equations C. Vuik, F.J. Vermolen, M.B. van Gijzen and M.J. Vuik Delft Academic Press, Delft, 2018 ISBN 978-90-6562-3737 The book is also online: https://textbooks.open.tudelft.nl/textbooks/catalog/book/57	
Assessment	Project assignments plus oral exam in period 4A. The project assignments are practical assignments which have to be completed in Python. The markings are a Pass or a Fail. All assignments have to be handed in via Brightspace assignments and have to be completed with a Pass. The oral exam will determine the end result for this course, and will be graded with a final grade rounded to halves or wholes. An opportunity to do a resit of the oral exam will be offered in period 4B. Information about exact instructions and deadlines for enrolling for a resit oral exam will be posted in the course Brightspace by the end of the exam week of 4A. Students who receive a failing grade on their project assignments may request an additional assignment which successful completion of within the deadline will allow for a passing grade to be received for the project assignment.	
Enrolment / Application	Enrollment in Brightspace is considered registration for all project assignments. Students must enroll in Brightspace for the course no later than the first week of the classes in periode 4A. Enrollment for the oral exam will be mandatory via a separate online list, which will be communicated via the Brightspace environment. Enrollment for the resit for the oral exam will be via email (instructions will also be posted via a Brightspace announcement).	
Elective	Yes	
Tags	Algorithmics Lineair Algebra Mathematics Modelling Numeric Methods Practicals Programming Software	
Location	TU Delft	

NB3026	Biomolecular Ultrasound	2.5
Responsible Instructor	Dr. D. Maresca	
Contact Hours / Week x/x/x/x	0/0/0/0/0/0/4/0	
Education Period	4 4A	
Start Education	4A	
Exam Period	4 4A 4B	
Course Language	English	
Course Contents	Biomedical Ultrasound elective course: 2.5 ECTS, 4th quarter. In this course, students will receive an introduction to imaging methods and biomolecular tools that allow ultrasound to visualize cellular functions such as gene expression. Specifically, the course will focus on the following content: 1. Genetically encoded acoustic biomolecules 2. Ultrasound imaging methods 3. Biomolecular ultrasound applications	
Study Goals	At the end of the course, student should be able to: 1. Explain the concept of biomolecular ultrasound imaging 2. Apply image ultrasound techniques to detect acoustic biomolecules 3. Discuss strengths and limitations of biomolecular ultrasound	
Education Method	The course will consist of 4 hours of education per week during 4 weeks and will be divided into two parts: lectures and laboratory work. 1. Lectures: 2 hrs of lectures per week 2. Laboratory work: 2 hrs of lab work per week at the Imaging Physics department	
Assessment	Oral exam	
Enrolment / Application	This course has limited enrollment. Enrollment will be available after November 1 via the Nanobiology Bachelor Brightspace page and is done on a first come, first served basis with priority for Nanobiology students. Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	

NB3030	Nanobiology Independent Research	0
Responsible Instructor	Dr. J. Pothof	
Course Coordinator	J.C. Colgrove	
Contact Hours / Week x/x/x/x	variable	
Education Period	Different, to be announced	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	interdisciplinary independent research	
Study Goals	At the end of the project, students will have learned to: - develop feasible research questions - applying innovative methods to the address these questions - communicate about scientific work in an effective way - write reports about their findings	
Education Method	In this course you carry out an independent research project with the help of a scientific advisor from the program. If you would like to do such a project, you may join an existing project (an example would be the Corona Research super project spring of 2020) or propose one to the Programme director. Activities could include literature review, modeling, and lab work, and must include scientific discussion and report writing.	
Assessment	Grading is Pass/Fail. You will be assessed on your research log of work done and your final written report.	
Enrolment / Application	Students wishing to do an independent project should discuss this with the programme coordinator. They will need to write a proposal outlining the project, their learning goals, supervision plan and outcomes. The supervisor must be a Nanobiology GreenList supervisor. This proposal must be submitted to the director of this course at least one month before desired date of beginning the project.	
Special Information	In special circumstances the program may begin a project and open registration to any students at any time. Students may not do an Independent research project and their BEP in the same research group or with the same supervisor. They may also also not do it in the same research group as an Honours Programme project.	
Studyload/Week	Credits are variable, students receive up to 1 ECTS per week based on at least 28 hours per week, with fractions thereof for less weekly effort. Credit requests with justification (log book and final report) must be sent to the responsible teacher who will assess the effort and register credits in Osiris.	

TBM301A	Writing a Bachelors Thesis in English	2.5
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. M.A. Swennen	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who are in the process of writing their Bachelors thesis. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly text analysis and writing assignments.	
Study Goals	The course is primarily designed for students working on their Bachelors thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure your thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from corrections, lectures, assignments and tutor feedback.	
Reader	Course materials will be provided digitally.	
Assessment	Continuous assessment. Participants are to submit a number of text analysis and writing assignments in which they demonstrate their writing skills and their understanding of text composition and academic English. Students that do not submit their weekly assignments may be asked to leave the course. An 85% attendance record is required. This means students may miss no more than one session (preferably none). Students will receive their course credits when all conditions are met.	
Elective	Yes	
Tags	Diverse English proficiency Project	

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